HCSA Rev F release package

The facility taken apart, every flight part drawn and in 3D, the test pieces with their shop drawings and STEP files, what the computer tests found, and what is still open.

For	Marco Cross, fabrication and feasibility review, Tuesday 17 November 2026, 10:00, and Rick Fleeter
From	Manos Theofanous and the HCSA team
Date	14 November 2026
Replaces	The Master Document of 27 October 2026 (Rev E)
Status	Working baseline. Nothing in it is released for manufacture; the test pieces are for quotation

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The sheets are A3 and follow the text of their part. Page numbers are on the text pages; the sheets carry their own title blocks.

0. Read this first

This is the Rev F release package for Marco Cross, for the call on Tuesday 17 November at 10:00, on Google Meet. It replaces the Master Document of 27 October, which described Rev E. It is shorter, and it leads with the things the call is about: the parts, how they are made, and the test pieces a shop has to quote.

What changed since the 27 October document

Rev E became Rev F through the eight decisions of 2 to 5 November below. On 10 November Rick signed the ones that needed him and agreed to move the hardpoint.

Date	Decision	Why
2 Nov	Secondary ribs I 180 x 60 x 10, web 5, a custom extrusion	Thinner ribs, more open glass; checked in the rib study
2 Nov	Corner tie preload 198 kN, held to 196.5 to 198.1 kN, was 203	The corner model with the real rib loads
4 Nov	Main rib option B, I 400 x 100 x 25, web 8, with bolted cap angles	3.8 t lighter than option A, with about 61 percent of each window open glass
4 Nov	Cabin at 0.8 atm with 26 percent oxygen; a module for a 1 atm commercial station, joined through a transfer lock	Takes a fifth of the load off the structure; the same oxygen pressure as at sea level
5 Nov	Pressure pane 21 mm BOROFLOAT 33, in 531 mm panes, 96 to a hexagon; was 22 mm	SCHOTT makes 21 mm as a standard 1,700 x 1,300 sheet; 22 mm would need container loads
5 Nov	Edge frame 180 mm deep; glands 7.1 x 4.0 mm; seam bolt NAS6705U27 with a KFN670 nut at 4 kN	The seam model at 180 mm passes with about 40 percent margin
5 Nov	Seam bolt row stays at z' 58	Moving it to z' 70, to give the seal loops rounder corners, failed the seam model
5 Nov	We stop quoting 35 MPa as the glass allowable until our own bars measure it	SCHOTT's figure is 25 MPa, a 5 percent fractile on abraded glass, with no Weibull data
10 Nov	Rick signs the 0.8 atm cabin, withdraws leak before break as a requirement, and confirms the factors, 1.5 at proof and 2.0 at ultimate	In the plain bays the metal yields before the seals open, so leak before break cannot hold; the factors are NASA's
10 Nov	The hardpoint fitting moves onto the line of the pull (z' 46 to 106, was 110 to 170)	As drawn the frames hinge about it and the seals open at 0.2 to 0.5 atm

The position of the hardpoint is settled, but the part behind it still has to be redesigned to fit the 180 mm frame (part 2, sheet RF-07).

What the computer tests found, in two sentences

At 0.8 atm the corners pass with about 12 kN of room on the worst tie, and still pass, by 0.7 kN, if the hatch is left open and the cabin goes to 1 atm; the seams pass with about 40 percent margin. The hardpoint as drawn does not work, because the fitting sits 64 mm deeper than the line of the pull and the two frames hinge about it, so the seals open at 0.2

to 0.5 atm; moved onto the line of the pull the seal at the fitting holds to 2.5 atm, with the bolt beside it 3 percent short of its requirement, and Rick agreed that move on 10 November.

How to read the package

- Part 1 is the facility as a whole, taken apart, with the spec card. Part 2 is every flight part, drawn and in 3D. Part 3 is the test pieces, with the drawings and STEP files a shop quotes from. Part 4 is what the computer tests say. Part 5 is everything still open, by subsystem. Part 6 is the questions for the call, in order. Part 7 says where every file lives.
- The sheets are A3 and sit behind the text of their part. Every sheet has a status line in its title block. LOCKED is spec, DECIDED is a decision Manos has taken, PROPOSAL is drawn to carry a load from the models but not yet tested or signed, OPEN is a decision not taken, with the default stated.
- Depths. On the panels, z runs from the outer face at 0 to the cabin seat plane at 180 mm, measured square to the panel. On the joint face of a seam, which is mitred, the same features sit at z', and the table on sheet RF-05 converts one to the other for the hexagon (times 0.934) and the pentagon (times 0.959).
- Every number in the text has a source in the project notes or the parameter file, and the sheets are drawn from that file. Where a number is a model result it says which run.

Words

Frame A and frame B are the two halves of a seam: A has the flat joint face, B has the glands. A gland is the groove the butyl cord sits in. The corner block is the machined node at each panel corner, and the ties are the bolts that hold three blocks together. The hardpoint is the pinned fitting along a seam that carries the pull across it, so the seam bolts only clamp the seals. Proof is 1.26 atm and ultimate 1.67 atm, on a design pressure of 12.3 psia with our own relief valve. The 550 article is the 550 mm long test piece of a seam with a dummy hardpoint; the T1 slices are the 75, 150 and 225 mm pieces pulled apart in the 100 kN frame.

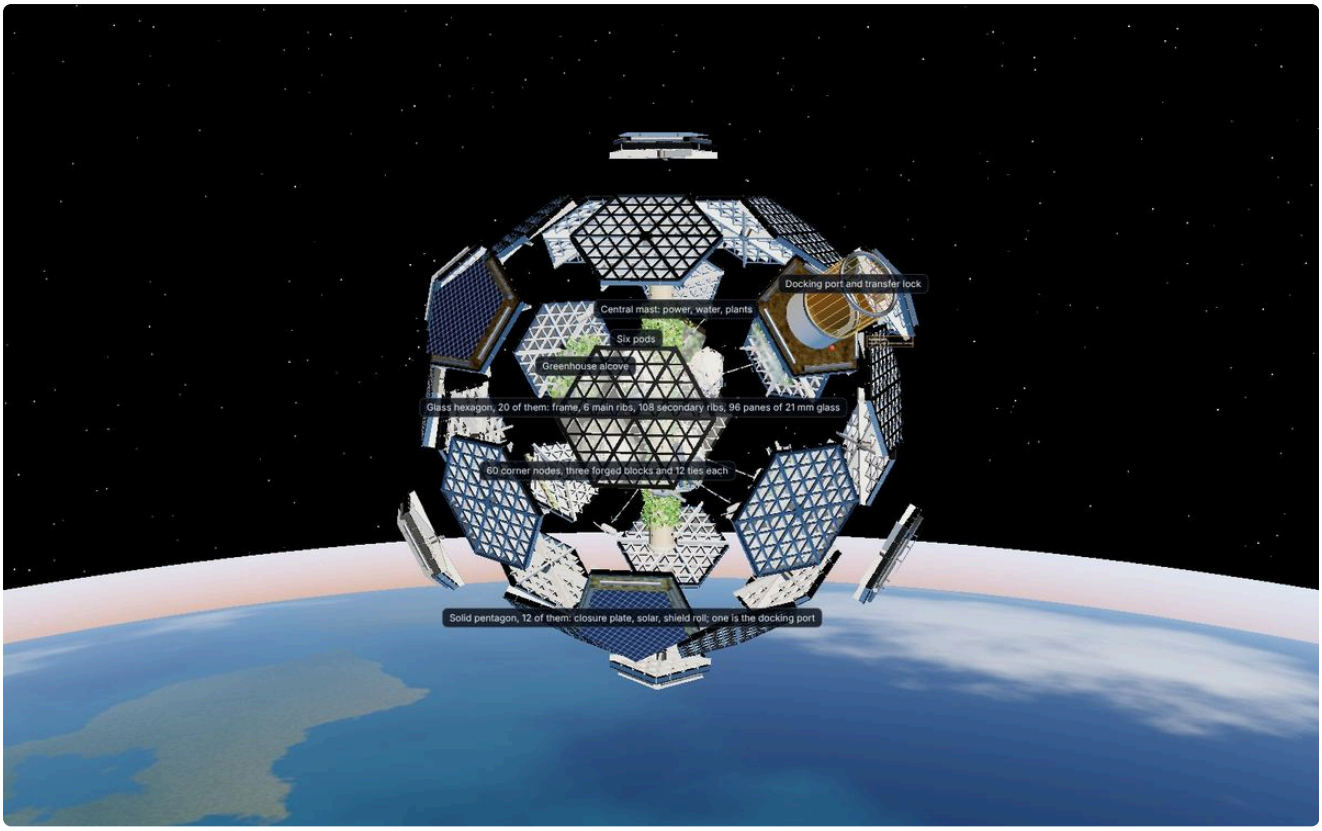
1. The facility

The spec card, Rev F

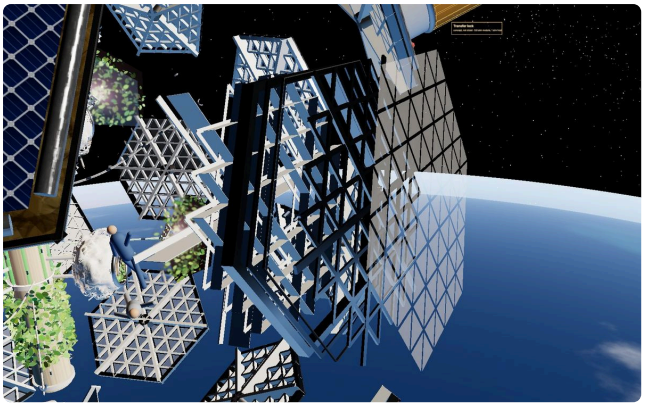
Item	Value	Status
Shape	Truncated icosahedron, 32 flat panels, 20 glass hexagons and 12 solid pentagons, 90 seams, 60 corners	LOCKED
Size	11,151 mm across the corners; every edge 2,250 mm	LOCKED
Cabin	0.8 atm (81 kPa) with 26 percent oxygen; design pressure 12.3 psia with our own relief valve	DECIDED 4 Nov, signed by Rick 10 Nov
Host	A module for a 1 atm commercial station, joined through a hatch-shut transfer lock	DECIDED 4 Nov
Factors	1.5 at proof (1.26 atm), 2.0 at ultimate (1.67 atm)	Confirmed by Rick 10 Nov
Main rib	Option B, I 400 x 100 x 25, web 8, 6 per hexagon and 5 per pentagon, with two bolted cap angles	DECIDED 4 Nov
Secondary rib	I 180 x 60 x 10, web 5, custom extrusion, on a k4 grid at 562.5 mm pitch	DECIDED 2 Nov
Edge frame	180 mm deep, 20 mm web, 50 x 10 flanges, seat lip at the cabin face; 6061-T651 for tests; the flight alloy is open (all checks are against 6061-T6)	Section DECIDED 5 Nov; alloy OPEN
Pressure pane	21 mm BOROFLOAT 33, 531 mm panes, 96 per hexagon, 1,920 on the shell, on 2 mm setting strips	DECIDED 5 Nov
Seals	Two loops of Parker BA500 butyl, 5.33 mm cord, in 7.1 x 4.0 mm glands at z' 18 and 32; RTV566 bead at z' 42	Glands DECIDED 5 Nov; curing the RTV on the ground is open (Q80)
Seam bolts	NAS6705U27 with KFN670 nut, A286, 4 kN by torque, 16 per seam at 75 mm pitch in bays of 3, 5, 5 and 3, row at z' 58; 1,440 on the shell	DECIDED 5 Nov
Corner node	One machined block per panel corner, 180 blocks in 60 nodes of three, 120 degrees on a hexagon and 108 on a pentagon; four 7/8-14 Inconel 718 ties per face at 198 kN, limit 227 kN	Block PROPOSAL, preload DECIDED 2 Nov
Hardpoints	3 per seam at the quarter points, 270 on the shell; a titanium lug in a two-cheek clevis with three 40 mm pins at 60 mm pitch	PROPOSAL; position on the line of the pull agreed 10 Nov; the part does not yet fit the 180 mm frame (RF-07)
Mass	36.1 t in the shell model. The parts lists of 8 Nov add up to more, and the two are not reconciled yet (part 2)	Open

The facility taken apart

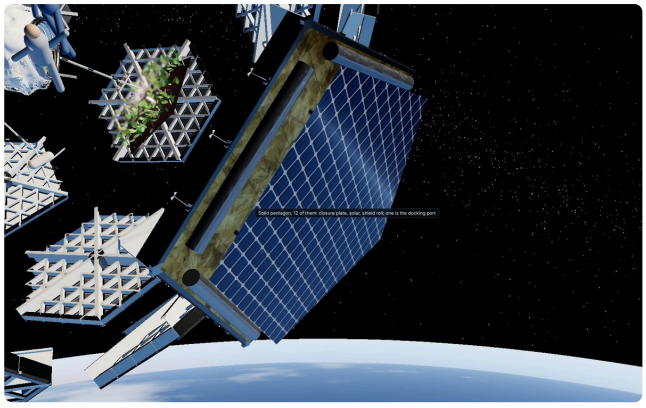
The team model at hcsa-site.vercel.app/model and the public page at /explore now have a taken-apart view. A slider moves the 32 panels out along their normals, lifts the corner blocks and the hardpoints off the seams, and sets the mast, the six pods and the pentagon kit apart with labels, so every piece in the parts list can be seen on its own and in its place. A second slider takes one hexagon apart into its frames, ribs, panes, blocks and hardpoints. The stills below come from that view.



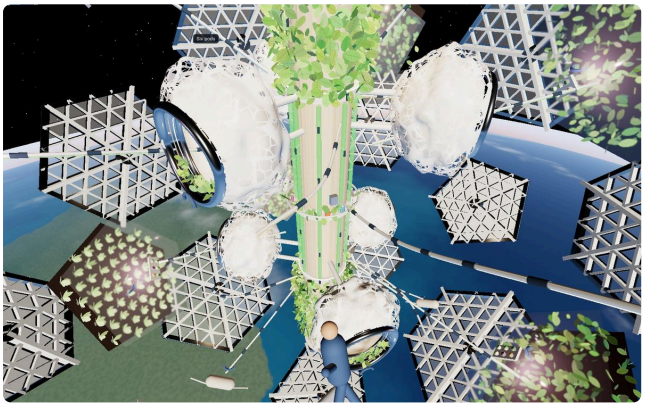
The facility taken apart: the 32 panels moved out along their normals, the mast, pods and greenhouse alcoves left in place, the docking pentagon with its transfer lock at the top right. From the master model, taken-apart view at 70 percent.



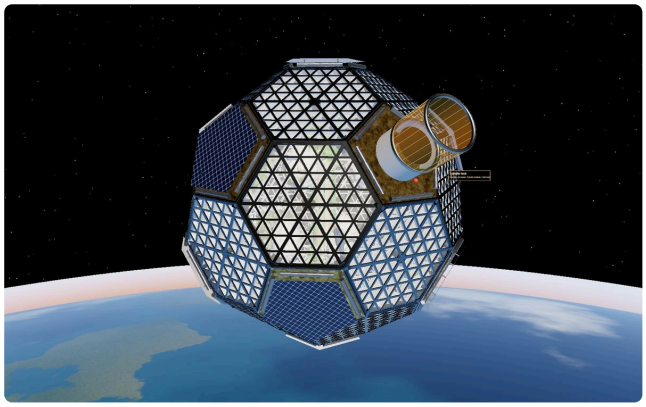
One hexagon opened into its layers: the frames, the main ribs with their cap angles, the secondary rib grid and the panes, each pulled out along the panel normal.



A solar pentagon with its blanket standing off the closure plate and the shield roll along one edge; the greenhouse alcoves show through the hexagons behind.



Inside, with the panels away: the mast down the middle, the six pods around it and the plant trays.



The same model assembled, for comparison.

What the pieces are, from the outside in:

- 20 hexagon panels, each about 1.5 t with its glass and before its blocks and hardpoints, and 12 pentagon panels. Every hexagon is the same part, and every pentagon is the same part.
- 180 corner blocks, one at every panel corner, tied in threes into 60 nodes.
- 270 hardpoint fittings along the seams, three to a seam, each a lug on one panel and a clevis on the other.
- 1,440 seam bolts, which only clamp the seals.
- The mast down the middle, with the six small pods around it, handrails and perches, and the plant trays.
- The pentagon kit: the closure plate, the solar blankets that stand off the structure, the fold-out shields, the dock and the transfer lock on one pentagon.

Links: the team model is at /model, the public one at /explore, the parts library at /parts and the sheets at /blueprints/#revf, all on hcsa-site.vercel.app. The team pages are behind the site password.

2. The flight parts, drawn and in 3D

Nine A3 sheets follow this page. They are drawn from the Rev F parameter file, not traced from CAD, and nobody has checked them for machining yet; that is what the call on 17 November is for. The same parts are in the parts library on the website, each with a STEP file, a GLB for the browser and four drawing views, and the whole facility is in the master model.

The sheets

Sheet	What it shows	Status
RF-01	The main rib, option B, and the secondary rib, cut through, with the 21 mm panes seated and the glass bite on each. The two new requirements are in the notes: the nodes must hold the main rib sideways, and the panes need an answer for launch	Ribs DECIDED; brace PROPOSAL; pane at launch OPEN
RF-02	A seam cut through at a plain bolt: the 180 mm frame, the two glands, the RTV bead, the bolt at 4 kN, and the study of the seal loop corners against Parker's R16 minimum	Frame, gland, bolt DECIDED; loop corners OPEN
RF-03	The T1 seam slices and the 550 mm article at 180 mm deep. Superseded in detail by the TP sheets in part 3, kept for the record	For reference
RF-04	The transfer lock between the 0.8 atm module and a 1 atm host, as a concept: the hatch that stays shut while the pressures differ, the equalising valves and our relief valve	Concept, not sized. Rick agreed the transfer lock with the 0.8 atm cabin on 10 Nov; the sheet's status line ("Rick to agree") predates that
RF-05	A hexagon-to-hexagon seam cut at the first bolt, 271 mm from the corner, with every feature on the joint face by z', both hardpoint positions as envelopes, and the table that turns z' into panel depth for the hexagon and the pentagon mitres	Seam DECIDED; hardpoint position OPEN; outer glass layers OPEN
RF-06	The hexagon panel from the cabin side: 6 main ribs, 108 secondary rib segments, 96 panes, 18 six-way nodes, 18 hardpoints, the 16 bolts on one edge, and the parts list with masses	Layout DECIDED; blocks, hardpoints and nodes PROPOSAL
RF-07	The hardpoint fitting as the October model has it, at the moved position: what fits (seals, bolts) and what does not (flange, ribs, pin insertion)	PROPOSAL, not yet buildable; position on the line of the pull agreed 10 Nov
RF-08	The corner node in plan and on the joint face, the C1 results, Marco's machining points (inside radii, a landing for the main rib, the material), and the P-11 brace	Block PROPOSAL; ties DECIDED
RF-09	The pentagon panel from the cabin side, with its roles, the closure plate as five triangles, and the solar stand-off	Ribs and frame DECIDED; closure side OPEN (Q57)

The parts library

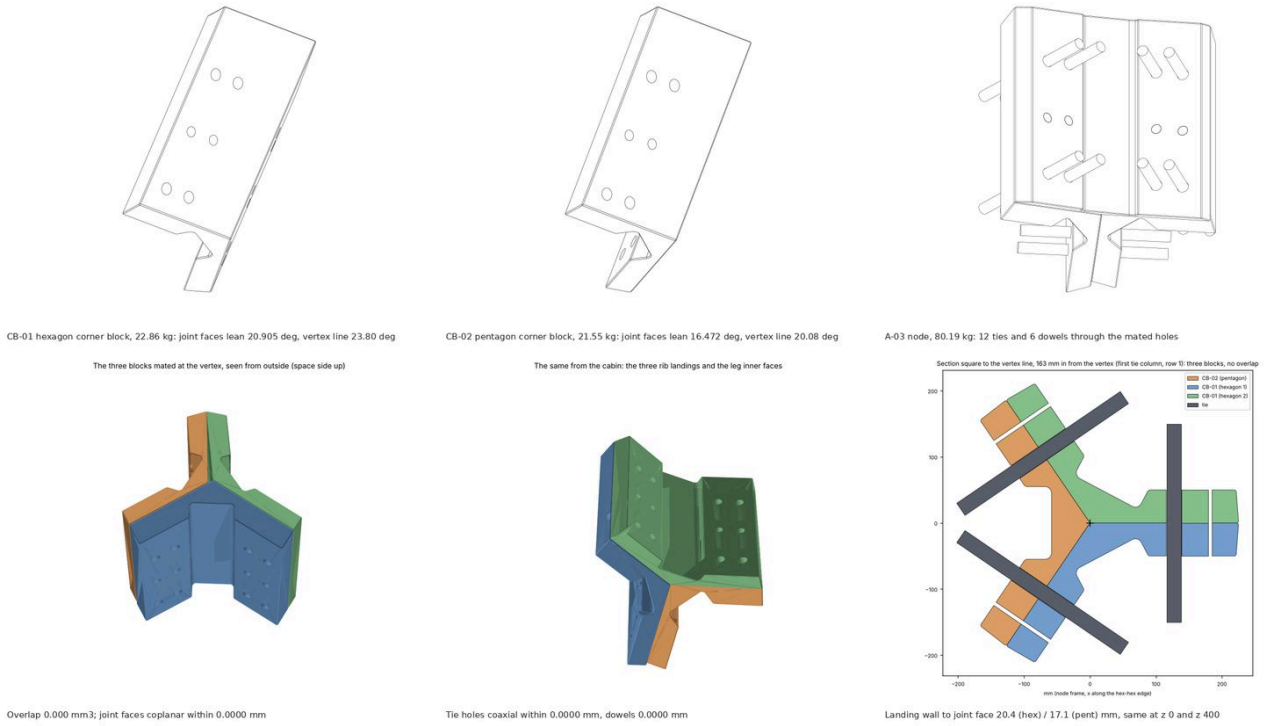
Every part below is at hcsa-site.vercel.app/parts, with STEP, GLB and four views. Masses are from the solids; the solids are weighed as 6061; the flight alloy is open.

Part	What it is	Each	Per hexagon	Per pentagon
P-06	Edge frame member, 180 deep, 2,250 long, with the glands, bolt holes and seat lip	27.9 kg	6	5
P-04	Main rib, option B, I 400 x 100 x 25, web 8	33.2 kg	6	5
P-04A	Main rib cap angle, one of the two that make the inner flange	12.8 kg	12	10
P-05	Hub core, 160 mm, with six forks for the main rib webs; the pentagon needs a five-fork version, not drawn	28.3 kg	1	1, five forks
P-05A	Hub star plate, 25 mm	14.2 kg	2	2
P-01	Secondary rib segment, I 180 x 60 x 10, web 5, 562.5 pitch	2.5 kg	108	90
P-02	Node star plate	1.8 kg	36	30
P-03	Node web core	4.4 kg	18	15
P-07	Pressure pane, 21 mm BOROFLOAT 33, 531 mm	5.5 kg	96	none
P-08	Pane corner clamp	0.2 kg	18	none
P-09	One seam bolt station, as an assembly		16 per seam	16 per seam
P-14	Seam bolt NAS6705U27 with KFN670 nut	0.03 kg	48	40
CB-01	Corner block, hexagon corner, with the real leaning joint faces; eight tie holes, dowel holes, a landing for the main rib, R10 inside corners	22.9 kg	6	none
CB-02	Corner block, pentagon corner	21.6 kg	none	5
P-06H, P-06P	Edge frame member with the web leaning 20.9 degrees (hexagon) or 16.5 (pentagon), 1,790 mm between the blocks; the straight P-06 stays as a schematic	22.5, 22.1 kg	6	5
HP-02A	Hardpoint lug body, Ti-6Al-4V, from the C3 model: a 39 mm plate, 200 wide, that crosses the joint	9.7 kg	9 (half of 18)	half of 15
HP-02B	Hardpoint clevis body, Ti-6Al-4V: two 22.5 mm cheeks	6.8 kg	9	half of 15
HP-03	Hardpoint pin, 40 x 95, A286, along the panel depth, in double shear	0.95 kg	3 per hardpoint	3 per hardpoint
A-05	The hardpoint assembled at the moved position, with the slot and the two plain bolts beside it	19.3 kg of fitting		
P-11	Main rib lateral brace at a node, 7075-T7351, as a pair	1.1 kg	18	15
PL-01	Pentagon closure plate, one of five triangles, 10 mm 6061-T651	44.9 kg	none	5
A-01	The node assembly: star plates, core, rib ends		18	15
A-02B	The hexagon panel assembly		1	
A-03	The corner node: two hexagon blocks and one pentagon block mated at a real vertex, 12 ties and 6 dowels	80.2 kg		
A-04	The pentagon panel assembly, schematic, still with the straight blocks of 8 Nov	922 kg		1

P-10 and A-02A, the option A rib and panel, are kept in the library as superseded. The shutters, the outer glass layers and the pentagon equipment are not in the library yet; they are not sized.

What the solids told us (8 and 9 November)

Building the parts as solids, and then checking them on the real polyhedron, is where most of this week's findings came from. An independent check of the whole package on 9 November found the rest. In order of what matters for the call:



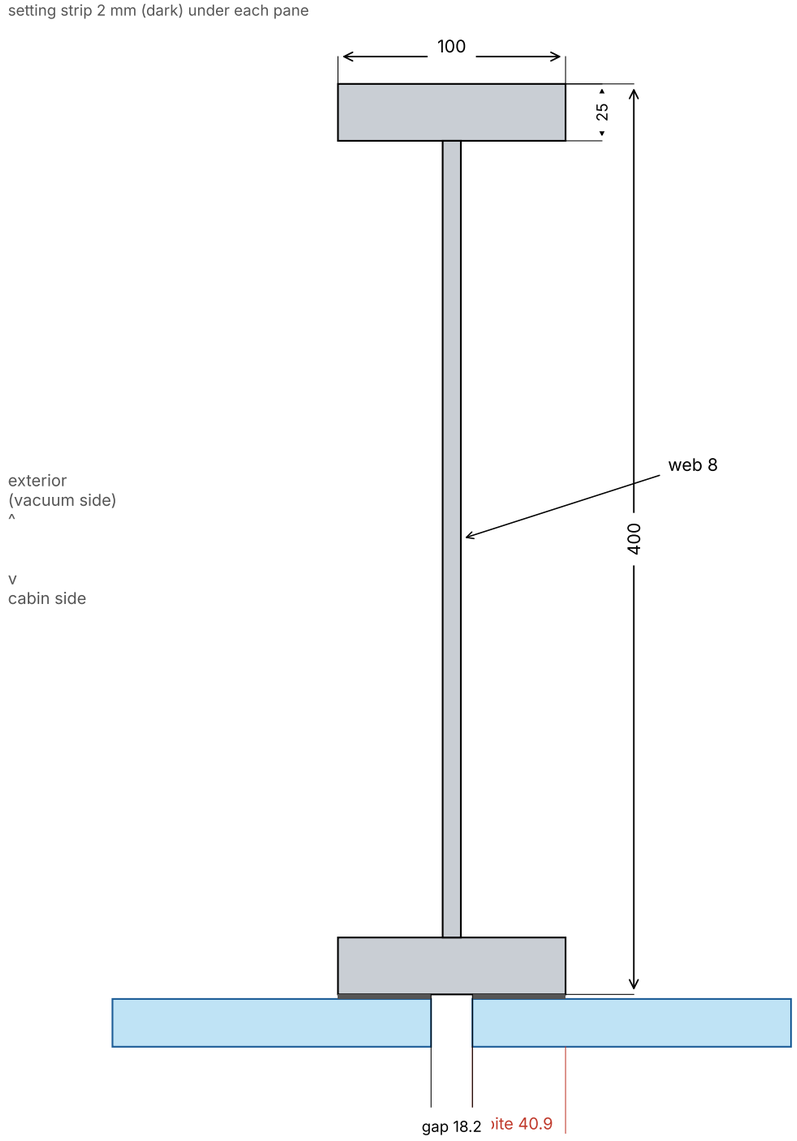
The corner blocks with their leaning joint faces, and two hexagon blocks and one pentagon block mated at a real vertex, from outside and from the cabin. Bottom right: a section through the first tie column, three blocks and no overlap.

1. **The corner node now fits.** On 8 November the blocks were drawn straight and three of them overlapped at a vertex. They are now one outline swept along the vertex line, with each joint face on the radial plane through its edge. Two hexagon blocks and one pentagon block placed at a real vertex share no volume, the joint faces are coplanar, and all 12 tie holes and 6 dowel holes meet on axis. The vertex line leans 23.8 degrees on a hexagon and 20.1 on a pentagon, so the main rib end has to be mitred to land on the block; P-04 is still drawn square.
2. **Tie columns at 130 and 186 mm from the vertex line.** The 87 and 143 mm in the corner model cannot clear the rib landing. On the pentagon block the washer seat is 3.4 mm from the landing edge, which is tight. C1 has to be rerun with the ties here.
3. **The landing leaves 20.4 mm of metal to the joint face on the hexagon block and 17.1 on the pentagon block**, the same at every depth. The R12 we promised at the vertex is taken up by the landing; the inside corners are R10, 430 mm long, and are cut from the open side, not plunged.
4. **The hardpoint position is right and the hardpoint part is not yet a design.** The fitting in the October model is a 39 mm titanium lug on one panel reaching into a two-

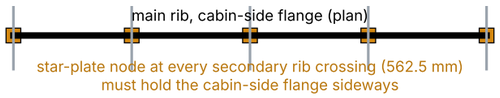
cheek clevis on the other, with three 40 mm pins along the panel depth in double shear. Our 8 November solids had the pins crossing the joint, which was wrong, and the clash we reported with the seam bolts came from a bolt drawn in the wrong place. With the real part at the moved position, only the lug crosses the joint, through a slot that is 10.5 mm from the RTV recess, 20 mm from the inner gland and clear of the seam bolts. That much works. What does not: the bodies had to move 38 mm apart to sit behind the 20 mm webs, so the lug is longer and the October stiffness no longer applies; the base blocks and pin ends run into the outer flange and into the two secondary ribs that land there; the pins cannot be driven in past the seat lip and the pane; and nothing says how the base blocks fasten to the frame. Sheet RF-07 shows all of it. The fitting has to be redrawn for the 180 mm frame and both models rerun.

5. **The frame is 180 mm deep square to the panel**, like the ribs, so the leaning joint face is 192.7 mm long on a hexagon and 187.7 on a pentagon. Sheet RF-05 is drawn 180 along the face and says so; the features on the face (glands, RTV, bolt row) are placed along it and do not move.
6. **The brace P-11** steps out 12 mm to pass the cap angle's leg; its size has no check behind it.
7. **The closure plate** is cut 10 mm back like the panes; the hub's fork cheeks cross its edge and need a notch. Which side it sits on is still Q57.
8. **Mass does not reconcile yet.** The shell model says 36.1 t. Adding up the parts lists gives about 30.5 t of hexagons with their 10.6 t of glass and about 9.8 t of pentagons, so about 40 t before the corner nodes (60 at 80 kg, 4.8 t, part of which is frame length already counted) and the hardpoint fittings (270 at 19.3 kg, 5.2 t). The solids are weighed at the density of 6061; in 2219 or 7050 the aluminium is 4 to 5 percent heavier. Whether the 36.1 t includes the glass and the fittings has to be traced in the model. Until then the honest figure is 36 to 50 t, and the nodes and the titanium fittings are the two places to look for weight.
9. **Naming.** On sheet RF-01 (5 Nov) P-04A meant the pentagon main rib. Since 8 Nov, P-04 is the main rib on both panels and P-04A is the cap angle.

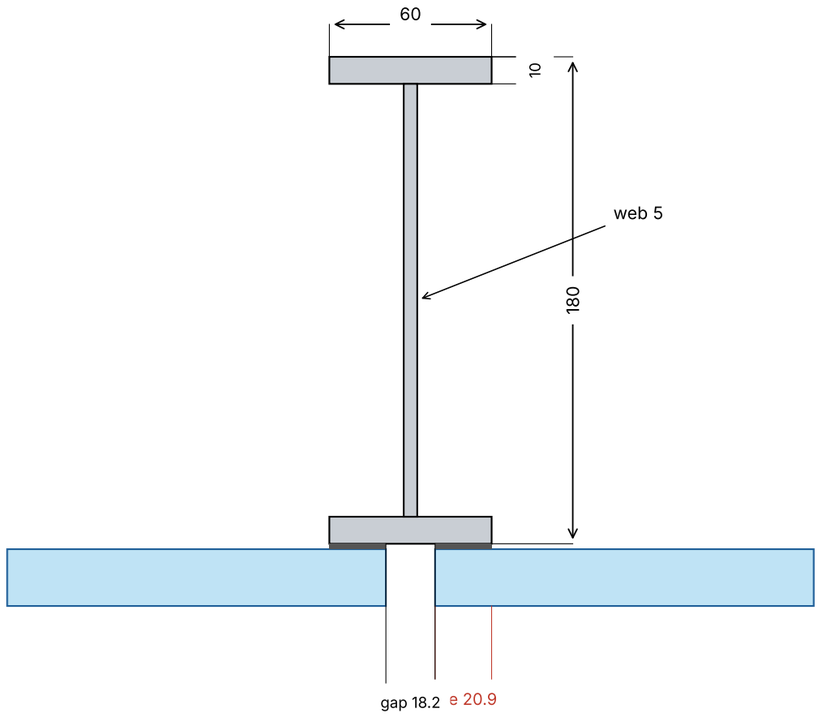
Main rib, option B: I 400 × 100 × 25, web 8
P-04 (hexagon), P-04A (pentagon), 6061-T6, 21.1 kg/m



21 mm BOROFLOAT 33 pane, 531 mm cut, cut 9.09 mm short of the rib centre line



Secondary rib: I 180 × 60 × 10, web 5
custom extrusion, 6061-T6, 5.4 kg/m

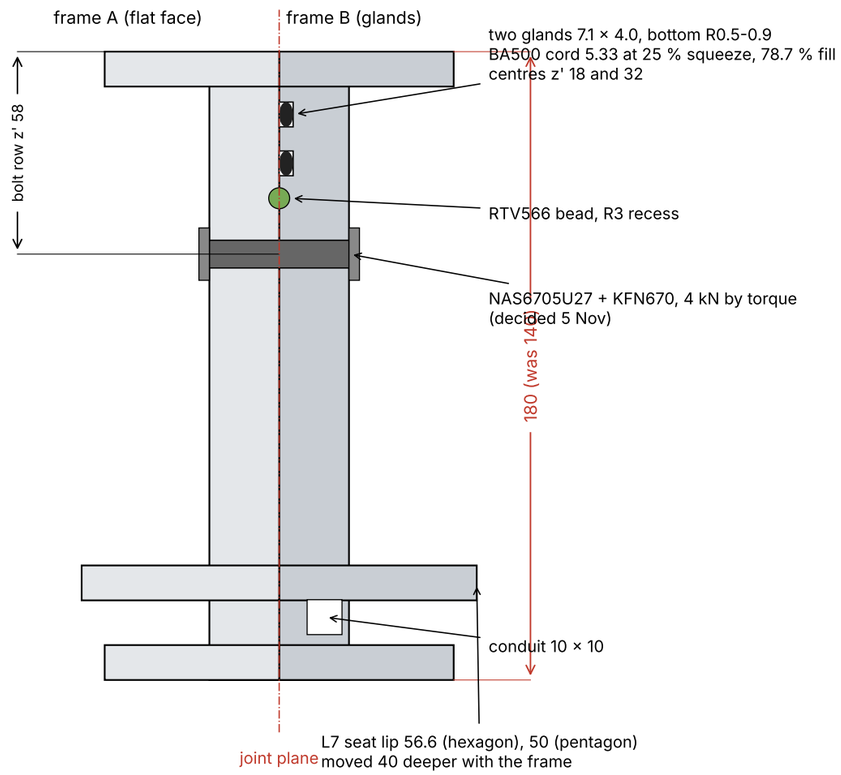


21 mm BOROFLOAT 33 pane, 531 mm cut, cut 9.09 mm short of the rib centre line

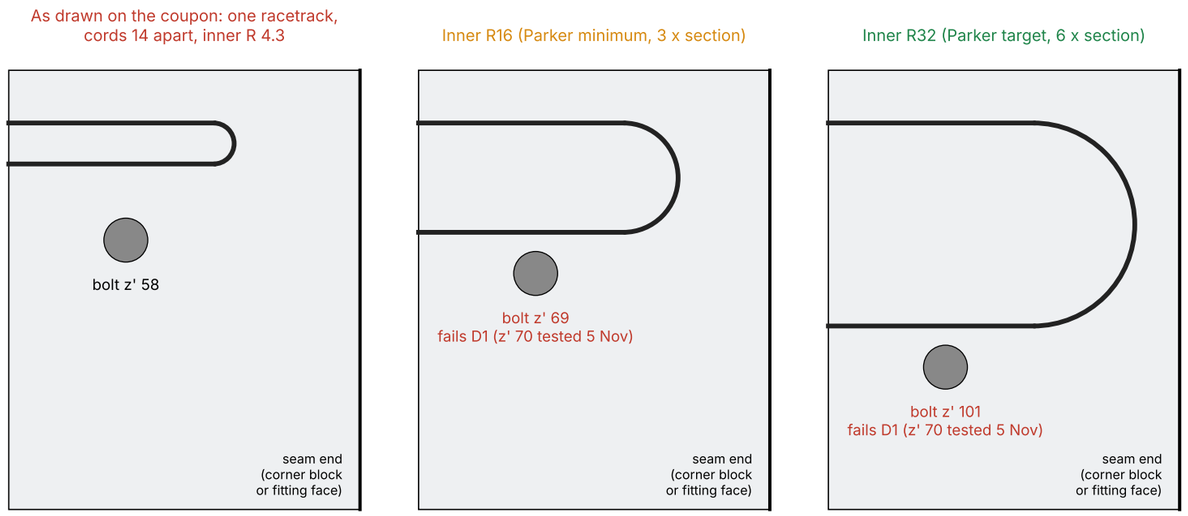
Rule checks: glass bite >= 20 mm on every seat (secondary 20.9, main 40.9); pane edge clears the web by 6.6 (secondary) and 5.1 mm (main).
Option A (I 450 × 150 × 25, web 10, parts P-10/P-10A) is superseded, kept for the record only.
Main rib B yields at 1.96 atm; at the 0.8 atm cabin the ultimate is 1.67 atm.
Open glass straight on with these ribs and a 60 mm frame band: 60.9 % of a hexagon (k 4 grid, exact drawing).
REQUIREMENT (survival check, 4 Nov): the main rib fails lateral-torsional buckling if unbraced over 2.25 m (0.79 of the load it must carry), and has 8.4 times margin if braced at 562 mm. The star-plate node at each secondary rib crossing must restrain the cabin-side flange sideways; carry this note onto RE-05 and RE-06.
LAUNCH NOTE: packed outer face down at 6 g, each pane pulls off its seat with about 0.35 kN. Pack the panels outer face up, or rate the pane clamps for 6 g plus random vibration (OPEN). Carry this onto the pane and seat sheet RE-04 and the packing plan (TRUTH 8.5.5).

HCSA RF-01 Main rib B and secondary rib with panes	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: main rib B 4 Nov, secondary 2 Nov, pane 21 mm 5 Nov: decided	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked

Section through a seam, normal to the joint plane (schematic, miter not shown)



Seal loop ends on the joint face (exterior at the top, z' down). Which layout we build waits on Parker's 'two loops' question.

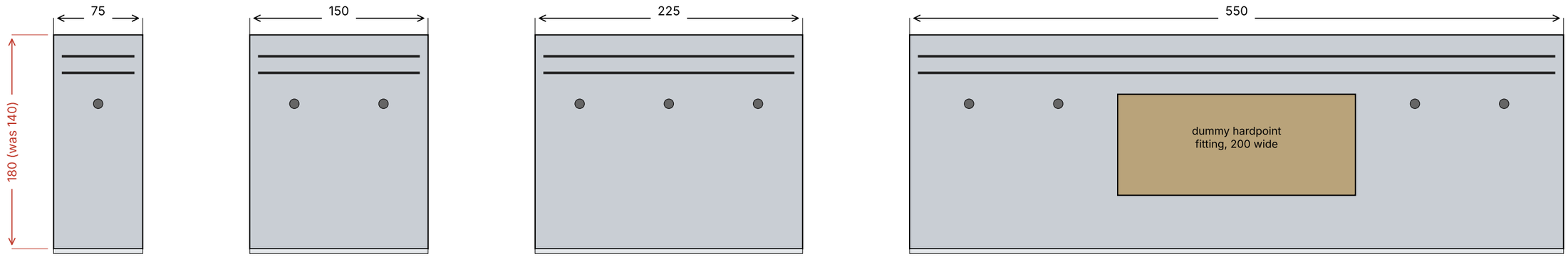


What the check says

- If each seam's two cords are one racetrack joined at the seam ends, the ends turn at inner R 4.3 mm. That breaks Parker's rule (R16 minimum).
- A racetrack with an inner R16 end needs its legs 37.3 mm apart, which pushes the bolt row from z' 58 to z' 70 or more (z' 102 for R32).
  The D1 run of 5 Nov says no: at z' 70 the seal gap is 0.111 mm at 2.48 kN per bolt, on the 0.11 limit (3.75 kN at z' 58). The bolt row stays at z' 58.
- If each cord is its own closed loop around the whole panel (Rev B intent), the turns are at the vertices, 60 deg on a hexagon, 72 deg on a pentagon,
  through the corner block and the O10 corner seal. That corner needs CAD of the block; it is not checked here.
- Next: draw the loop path through the corner block in Fusion
  (turn the loop round the end bolt, or widen the frame flange locally at the panel corners), after Parker answers 'two loops'.

HCSA RF-02 Seam frame 180, gland, seal loop corners	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: frame 180, gland 7.1, bolt 4 kN: decided 5 Nov; loop corners open	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked

Elevation on the joint face: frame depth 180, two cord lines at z' 18 and 32, bolt row at z' 58 (kept: z' 70 failed D1, 5 Nov)



RE-01-75
1 bolt at 75 pitch

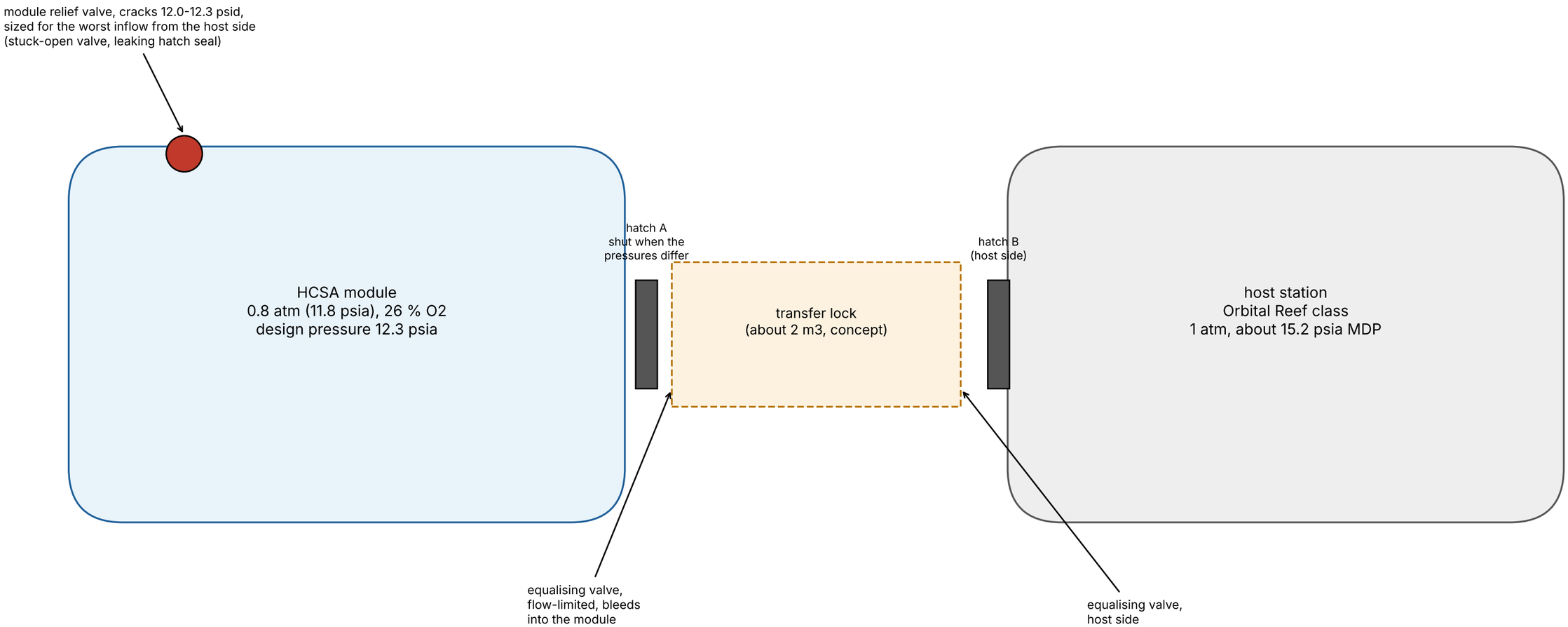
RE-01-150
2 bolts at 75 pitch

RE-01-225
3 bolts at 75 pitch

RE-02-550 with a dummy hardpoint
4 bolts at 75 pitch

Changes from the Rev E draft sheets RE-01 and RE-02:
- Frame depth 140 → 180 on every T1 frame, so the test matches the station frame (decided 5 Nov). Stock: 6061-T651 plate; check the blank with the shop.
- Glands 7.1 × 4.0 (decided 5 Nov). Cord ends at the slice ends follow RF-02: no tight racetrack end; use R16 minimum or dam the ends.
- Bolts NAS6705U27 + KFN670 at 4 kN by torque (decided 5 Nov), row at z' 58.
- The 550 article carries the dummy hardpoint of RE-03 rev B (no bolt inside the fitting: 2 + 2 bolts, 31.25 mm off its faces), pulled through pinned lugs.
- The pull blocks of RE-03 rev B keep their interface; their bolt holes move with the 40 mm deeper frame.
- The Instron adapter waits on Jerry's measurements.

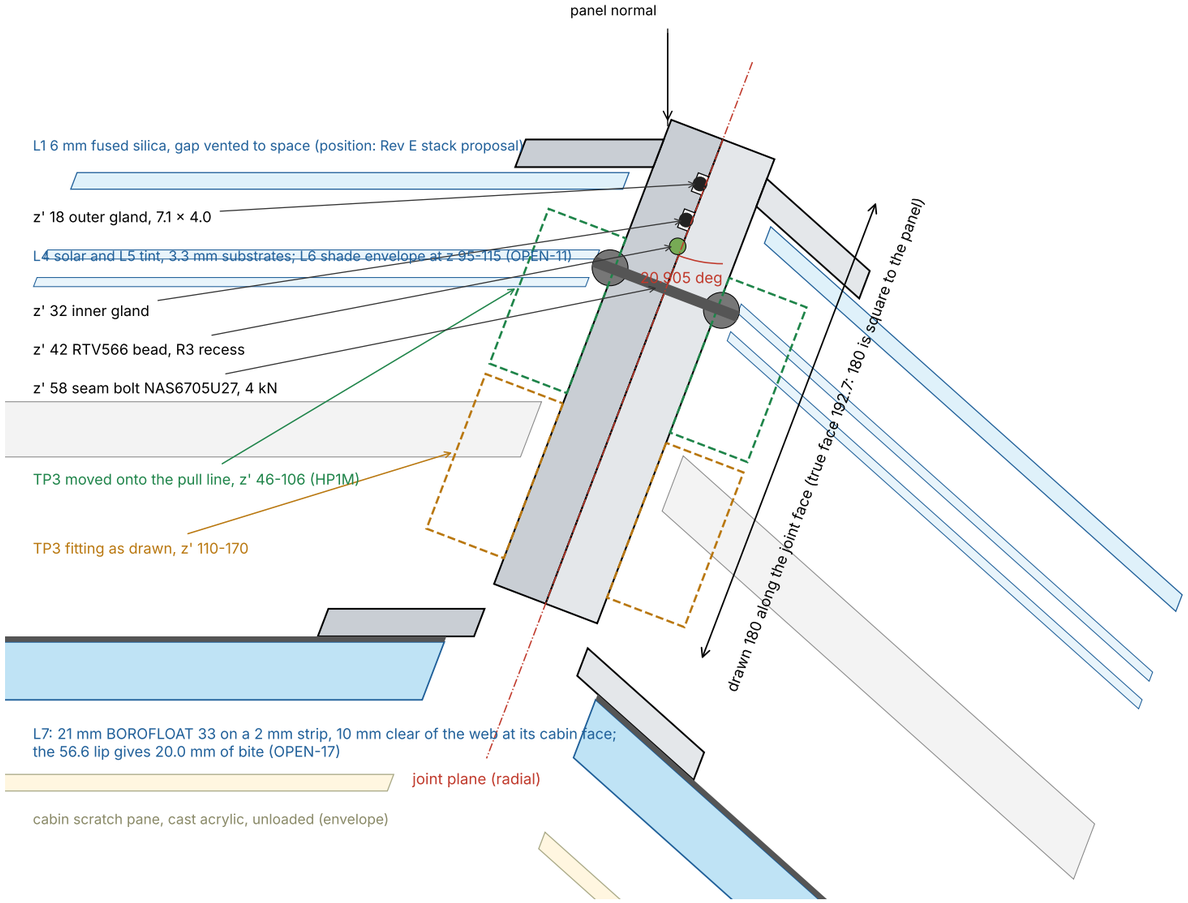
HCSA RF-03 T1 test articles at 180 deep	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: depth 180, gland 7.1, bolt 4 kN: decided 5 Nov	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked



Rules this concept has to keep (from the 3 Nov lower-pressure note):
1. The module has its own relief valve, cracking at about 12.0-12.3 psid, big enough for the worst inflow from the 1 atm side.
2. Hatch A stays shut whenever the pressures differ, held by the pressure difference and by an interlock.
3. The lock does the stepping through flow-limited valves: about 6-10 minutes at 0.3-0.5 psi per minute, slowest on the way back up to 1 atm.
4. The module never takes its air from the host. Volumes, valve sizes and the hatch are not sized; the operator has to agree to a hatch-shut interface.

HCSA RF-04 Docking transfer lock, concept	
Rev F working baseline, 5 Nov 2026 (supersedes the Rev E draft sheet)	Dimensions in mm
Status: concept, Rick to agree	Redrawn from hcsa_params_revF.json
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining until checked

Section normal to a hexagon-to-hexagon seam at the first bolt (271.25 mm from the vertex). Space side at the top. Frame B (left) carries the glands.



Joint-face features, placed along the joint plane (z')

Panel depth = z' x cos(mitre): hexagon 0.9342, pentagon 0.9590. Both frames of a seam share the plane, so holes and glands line up on hexagon-to-pentagon seams too (OPEN-9).

Feature	z'	Hexagon z	Pentagon z
Outer gland	18	16.82	17.26
Leak port	25	23.35	23.97
Inner gland	32	29.89	30.69
RTV recess	42	39.24	40.28
Seam bolt row	58	54.18	55.62
TP3 pads, as drawn	110-170	102.8-158.8	105.5-163.0
TP3 pads, moved (HP1M)	46-106	43.0-99.0	44.1-101.6
Corner tie rows	133.8, 380.0	125.0, 355.0	128.3, 364.4

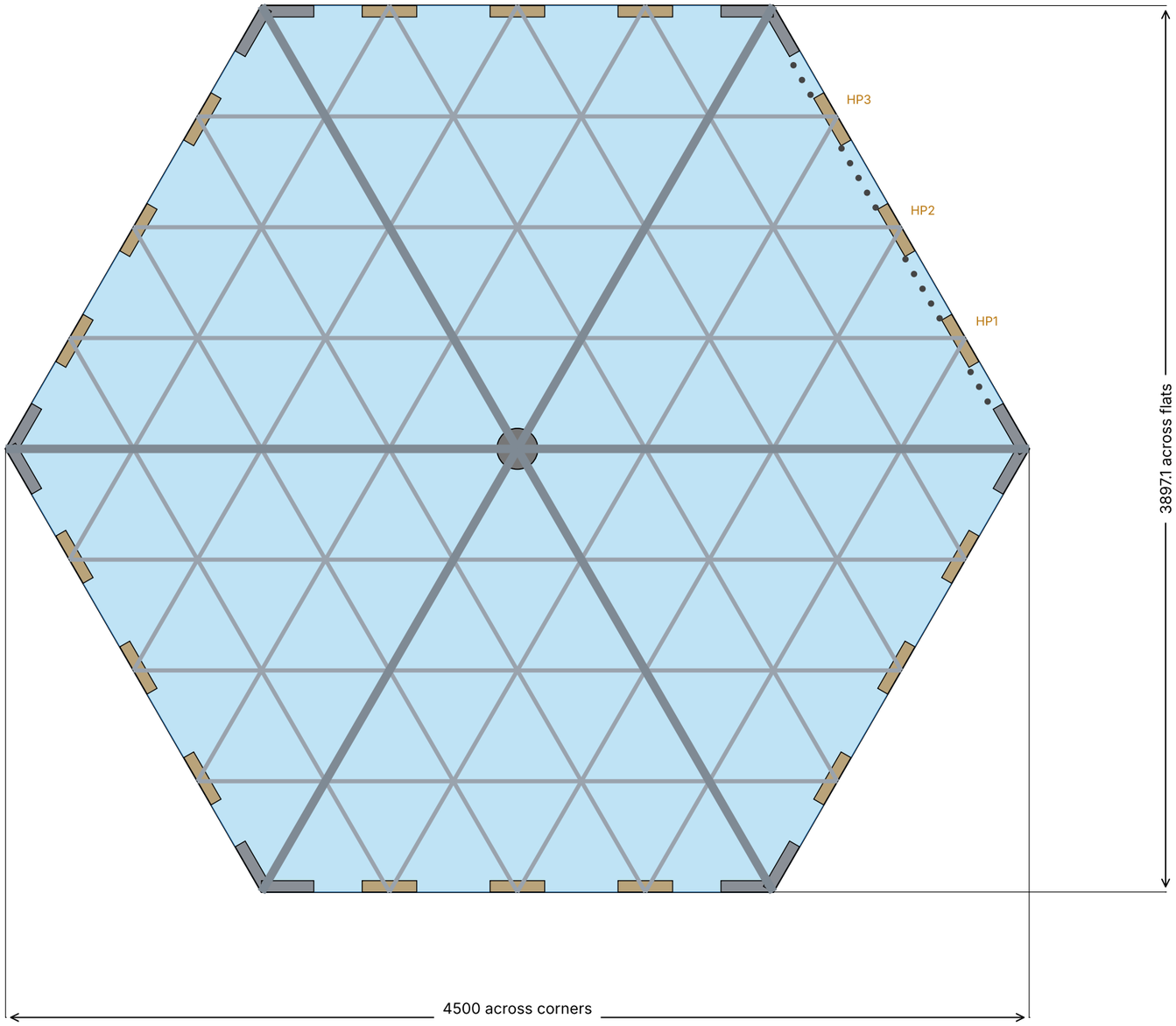
Seal: two loops of Parker BA500 butyl 5.33 mm in 7.1 × 4.0 glands (25 % squeeze, 79 % fill), RTV566 bead cured on the gro
Bolt: NAS6705U27 A286 with KFN670 nut, 4 kN by torque, 16 per seam at 75 pitch in bays of 3, 5, 5, 3.
Loop corners: R16 minimum at the corner block, OPEN (Q92).

What changed from RE-04 (24 Oct): frame 140 to 180; pane 22 to 21 mm; bolt NAS6705U27 at 4 kN (was NAS9306C-06, a rivet standard); the lip and seat moved to the cabin face (z 170-180, P-06) for the restack (Q78); the hardpoint shown in both positions because the fitting as drawn fails the HP1 seal check (0.22 atm free) and the moved fitting holds (0.064 mm at 1.93 atm); Rick decides on 10 Nov. Shear along the seam goes to the hardpoint pins and the vertex dowels, no shear key (OPEN-15, confirmed by FEA 24 Oct). Secondary ribs (I 180 × 60 × 10) and the main rib cap angles (z 375-400) are off this section: see RF-01 and RF-06.

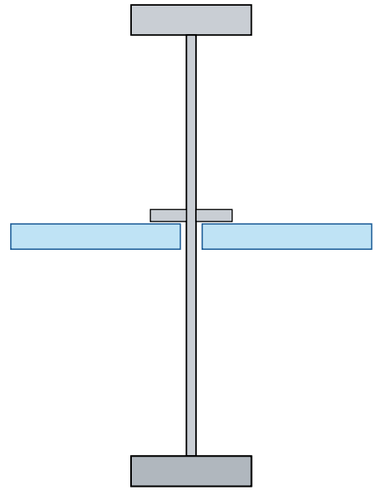
HCSA RF-05 Seam cross-section, hexagon to hexagon, at the first bolt	
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: frame 180, gland, bolt decided 5 Nov; hardpoint position open; outer layers OPEN-11	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Hexagon from the cabin side (schematic, 1:20 on A3): frames, six main ribs, the k 4 secondary grid, hub, corner blocks (grey), hardpoint blocks (gold), 96 panes

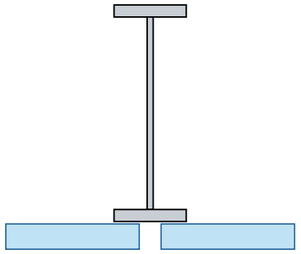
16 seam bolts per edge at 75 pitch in bays of 3, 5, 5, 3 (dots on the upper right edge); the first bolt 271.25 from the vertex; hardpoints HP at 562.5, 1125, 1687.5



Main rib B: I 400 × 100 × 25, web 8
seat ledges at z 170-180, cap angles P-04A at z 375-400



Secondary rib: I 180 × 60 × 10, web 5
pane seat on the cabin face z 180



21 mm panes on 2 mm setting strips, cut 9.09 mm short of the rib centre lines; bite 40.9 (main) and 20.9 (secondary); P-11 brace at every crossing (RF-08 r

k 4 grid: 562.5 mm pitch, 108 secondary rib segments, 96 panes (48 full triangles, 48 cut back at the frame and the hub), 18 six-way nodes, 18 crossings on the main ribs, 18 hardpoints

Parts list, one hexagon (model masses)

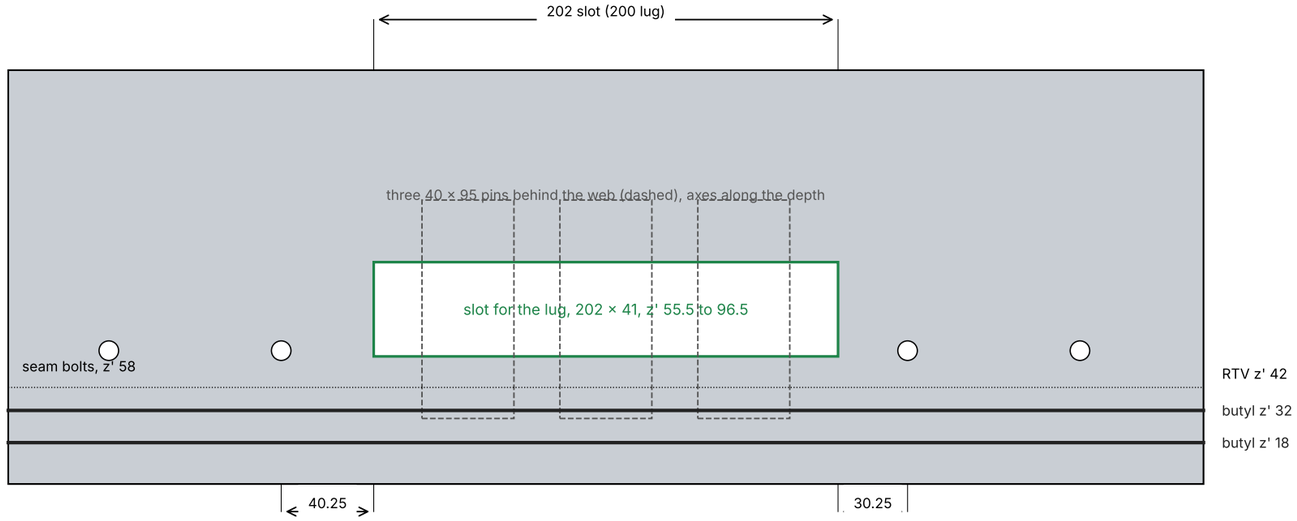
Part	Qty	Each kg	Total kg
P-06H frame member, 1,790 between the blocks	6	22.55	135
P-04 main rib B	6	33.24	199
P-04A cap angle	12	12.77	153
P-05 hub core	1	28.28	28
P-05A hub star plate	2	14.20	28
P-01 secondary rib segment	108	2.53	273
P-02 node star plate	36	1.75	63
P-03 node web core	18	4.38	79
P-07 pane, 21 mm	96	5.50	528
P-08 pane corner clamp	18	0.19	3
P-14 seam bolt (half of 16 × 6)	48	0.03	1
CB-01 corner block, leaning faces (9 Nov)	6	22.86	137
HP-02A lug body, Ti-6Al-4V (half of 18 hardpoints)	9	9.69	87
HP-02B clevis body, Ti-6Al-4V (the other half)	9	6.77	61
HP-03 pin 40 × 95, A286 (3 per hardpoint, shared)	27	0.95	26
P-11 main rib brace pair, 7075	18	1.12	20

Listed parts: 1824 kg per hexagon with its glass (528 kg) and its half share of the hardpoints. Masses at 6061 density; shell model 36.1

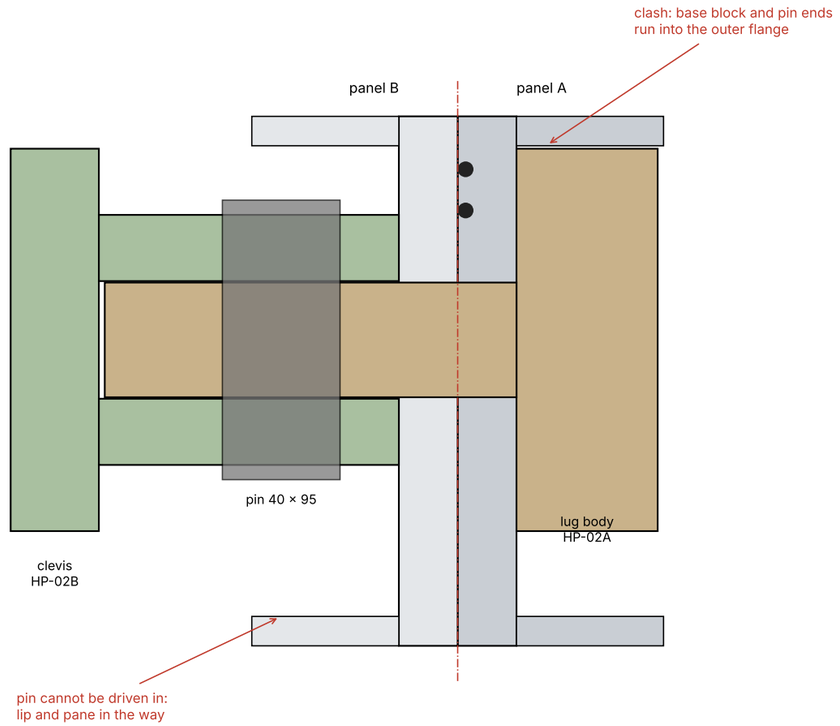
HCSA RF-06 Hexagon panel, general arrangement and parts list

Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: ribs and frame decided; blocks, hardpoints, nodes PROPOSAL; model masses	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Joint face at a hardpoint (HP1 of a seam): only the lug crosses the joint, through a slot on the cabin side of all three seal lines



Section across the seam, drawn straight (space side at the top): lug in two cheeks, pins in double shear



What the fitting is. Measured from the October C3 model (the run that gave 2.876 million N/mm): a Ti-6Al-4V lug 39 thick and 200 wide on panel A reaches across the joint into a clevis of two 22.5 mm cheeks on panel B; three A286 pins, 40 × 95, at 60 pitch, run along the panel depth with no clearance and work in double shear. The 8 November solids (pins across the joint, a pad of 200 × 60) were wrong and are withdrawn.

What the models say (HP1, 6-7 Nov). With the fitting where sheet RE-06-1 put it, 64 mm deeper than the line of the pull, the frames hinge about it and the seal gap reaches 0.11 mm at 0.22 atm (0.47 with the frames held) against the 1.93 atm required. With it on the line of the pull (lug mid-plane at z' 76) the gap at the hardpoint is 0.064 mm at 1.93 atm; the band at the plain bolt beside it reaches 0.11 mm at 1.87 atm, 3 % short. Runs with that bolt closer (115 mm) and tighter (6 kN) are on the laptop, 9 Nov. Those runs model the fitting as a spring between rigid pads of 200 × 60, not this part.

What the solid found (9 Nov, leaning hexagon frame). Good: the slot is 10.5 mm from the RTV recess, 20 mm from the inner gland and 32 mm from the nearest bolt hole; the seam bolts on the flight layout (131.25 and 141.25 mm from the hardpoint centre) are clear of it; the 8 Nov 'pad covers the bolts' clash was a bolt drawn in the wrong place. Not good: (1) to sit behind the 20 mm webs the bodies had to move apart 38 mm, so the lug is 140 long, not 102, and the pin line is 60 mm from the joint, not 40: the C3 stiffness no longer applies. (2) The base blocks and pin ends run into the outer flange, and both bodies and the pins run into the two secondary ribs that land on the hardpoint. (3) The pins cannot be driven in: the seat lip and the pane are in the way from the cabin side, the flange from the space side. (4) The web is cut through over 202 mm at the bolt row's depth, and nothing yet says how the base blocks fasten to the frame. The old 50 mm local block (HP-01) has no job left. So the position is right and the part is not yet a design: it needs to be redrawn for the 180 mm frame (a shallower stack, pins put in from the side or along the seam, the ribs landing on the base blocks) and C3 and HP1 rerun on that. Load 108.7 kN per atm on the worst hardpoint; one hardpoint is 19.3 kg.

The three hardpoints on a 2,250 mm seam, from the vertex, with the nearest seam bolts (bays 3-5-5-3 at 75 pitch)

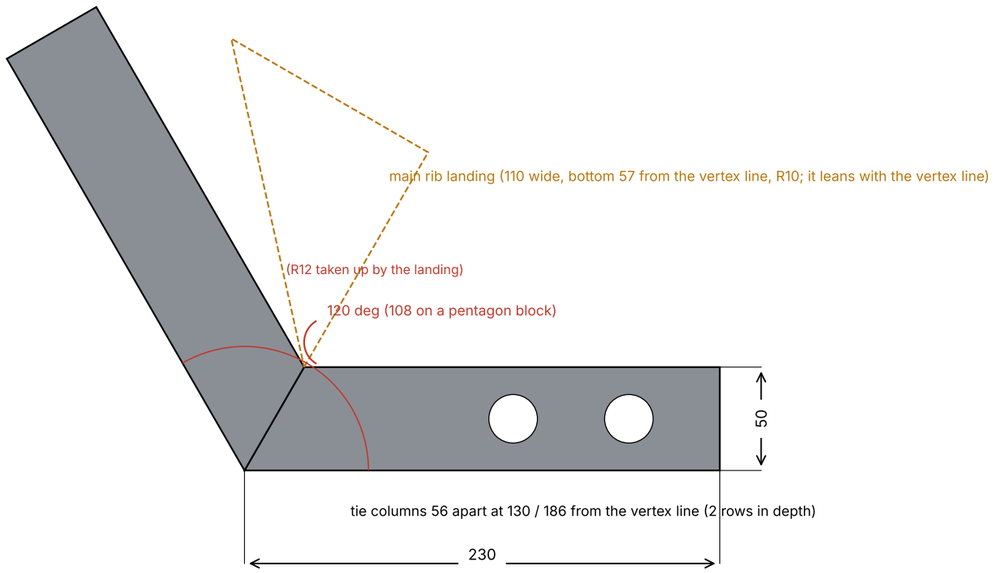
HP1 at 562.5: slot from 461.5 to 663.5; nearest bolts 421.25 (141.25 from the centre) and 693.75 (131.25 from the centre)

HP2 at 1125.0: slot from 1024.0 to 1226.0; nearest bolts 993.75 (131.25 from the centre) and 1256.25 (131.25 from the centre)

HP3 at 1687.5: slot from 1586.5 to 1788.5; nearest bolts 1556.25 (131.25 from the centre) and 1828.75 (141.25 from the centre)

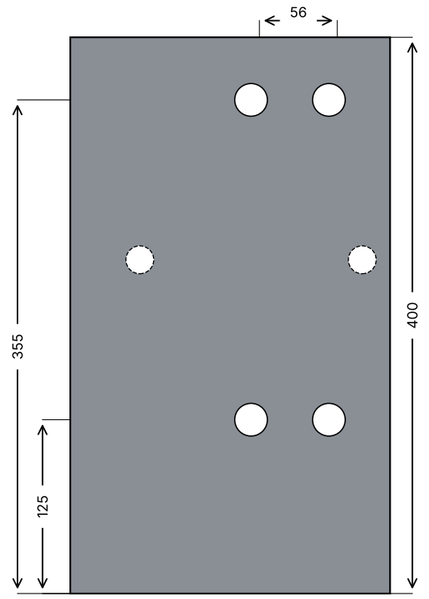
HCSA RF-07 Seam hardpoint: the TP3 fitting from the C3 model, at the moved position	
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: PROPOSAL, not yet buildable; position decided Tue 10 Nov	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Plan of one hexagon corner block (CB-01), looking at the cabin face



joint faces: the legs' outer faces, on the radial joint planes

Joint face of a leg (z' 133.8 and 380 along the plane)



4 ties 7/8-14 Inconel 718 (23.5 holes), 2 dowels 20 (24 room)
hexagon rows z 125 / 355, pentagon rows z 128.3 / 364.4

What the corner model says (C1, 5 Nov)

- Tie preload 198 kN, held to 196.5-198.1 kN, by stretch
- Worst tie at proof, 0.8 atm: 214.5 kN (limit 227)
- Pentagon 15 K warmer: 213.3 kN
- Hatch open, 1 atm: 226.3 kN, 0.7 kN under the limit
- Faces open at proof: 0.55 mm (0.74 at 1 atm), limit 0.8
- Block stress, 95th percentile: 236 MPa (255 at 1 atm), no limit set yet
- Worst corner load 235 kN at 0.8 atm, 290 at 1 atm

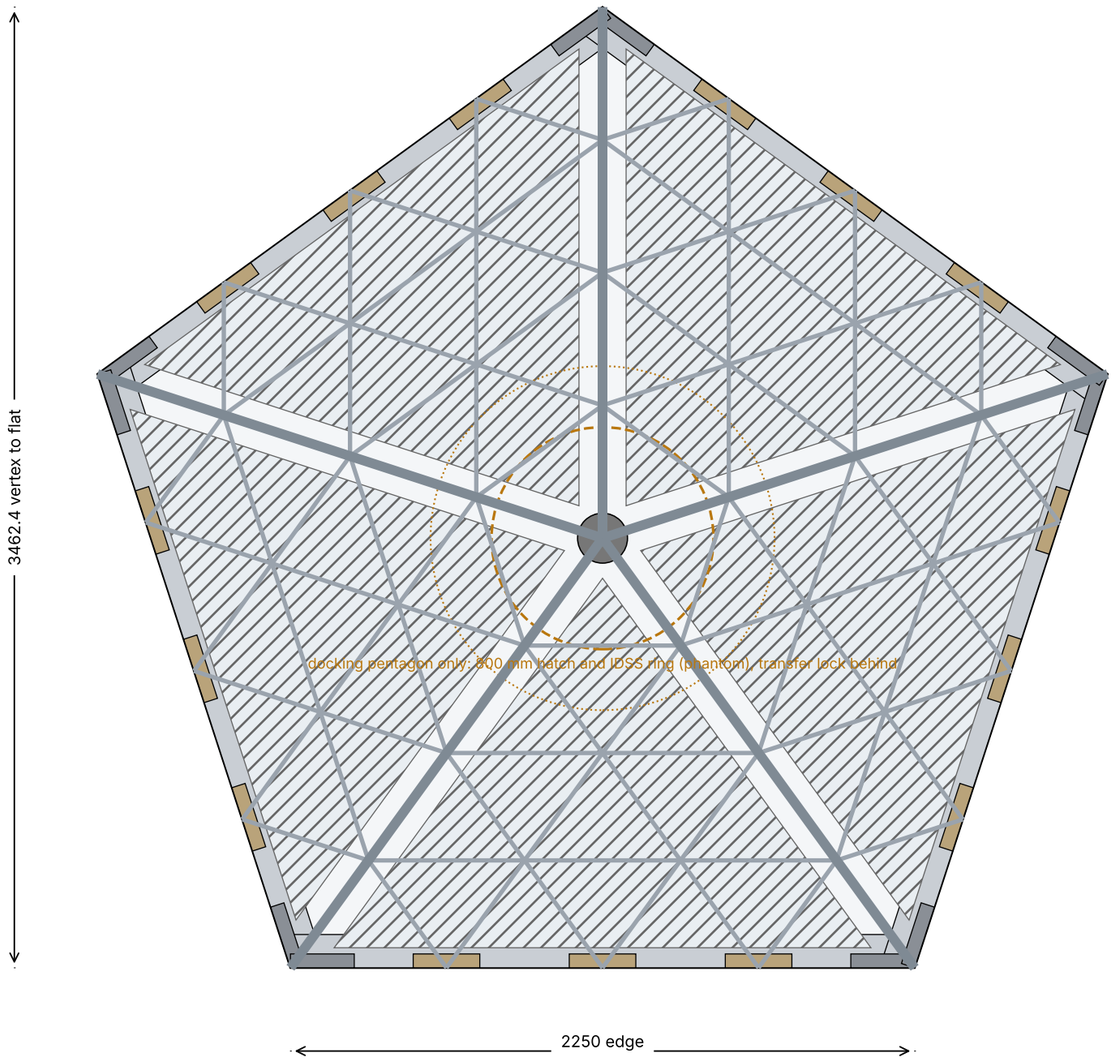
Making it (Marco, 7 Nov). As drawn the node cannot be machined: every inside corner needs a radius a cutter can reach (R10 here; the landing takes up the re-entrant corner), and the part could be cast instead of machined. Our position (9 Nov): machine it from 7050-T7451 or 2219-T851 plate, not a casting (seal faces) and not a die forging (180 parts). The main rib's flanges enter the block at the vertex landing (notch, PROPOSAL); the two frame members run into the legs' inner faces; the tie-to-web-fillet clash (Q29), the O10 maze-patch corner seal and the loop corners R16 (Q27, Q92) are open. The ties are tightened by measured stretch, because a torque wrench scatters about 25 % and the hatch-open case leaves only 0.7 kN. Forces Marco called 'achievable but unreasonable': 198 kN per tie (about 650-900 N m if torqued), 12 ties per node, 720 on the station. The corner node assembly A-03 (two hexagon blocks and one pentagon block at a real vertex, dihedral 138.19 / 142.62 deg) is in the parts library. Solids of 9 Nov, with the real leaning joint faces (CB-01 22.9 kg, CB-02 21.6 kg, node A-03 80.2 kg): three blocks placed at a real vertex share no volume, the joint faces are coplanar and the 12 tie holes and 6 dowel holes meet on axis. The block is one outline swept along the vertex line, which leans 23.8 deg (20.1 pentagon), so the main rib end is mitred to match. Tie columns 130 / 186 mm from the vertex line (87 / 143 cannot clear the landing). Metal from landing to joint face 20.4 mm (17.1 pentagon); washer seat to landing edge 14.3 mm (3.4 pentagon, tight). The R10 corners are 430 mm long: cut from the open side.

Main rib lateral brace (new requirement, 4 Nov): the main rib buckles sideways if unbraced over 2.25 m (0.79 of its load) and has 8.4 times margin braced every 562.5 mm. P-11 (PROPOSAL): a pair of 8 mm 7075 plates on the main rib web from z 180 to z 375 at every secondary rib crossing, bolted through the web, tying the cabin flange to the grid. Size to be checked.

HCSA RF-08 Corner node: blocks, ties, dowels, what the model says, machining

Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: PROPOSAL; ties, preload decided; radii per Marco; O10 seal, rib landing open	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Pentagon from the cabin side (schematic, 1:20 on A3): the same frame, main rib and secondary rib parts as the hexagon, no glass, a 10 mm closure plate



k 4 grid: 90 secondary rib segments, 5 main ribs, 5 corner blocks, 15 hardpoints; closure plate 10 mm 6061-T6 as five triangles (hatched), sealed at the main ribs

What the twelve pentagons do

Set by Manos on 25 Oct (PROPOSAL for TRUTH; the model and the earlier TRUTH split still show shield, service, docking and greenhouse pentagons):

- Eleven identical pentagons: inside, a greenhouse alcove and a service panel connected to the central mast; outside, solar panels on standoffs over an insulating blanket, and a stowed inflatable shield that cuts UV, takes heavy MMOD strikes and can act as a repair balloon (Q89).

- One docking pentagon: IDSS ring, 800 mm hatch, the transfer lock and relief valve (RF-04); the hatch stays shut whenever the pressures differ.

- Stabilising thrusters, if ever added, live in the pentagons (Q90). Which way the mast runs is open (Q88).

- The closure plate carries pressure only; a thin stressed skin would see 190 to 680 MPa. At 3.6 m across it is wider than plate comes, so it is five triangles sealed at the main ribs. Which side of the ribs, and how it is held, is OPEN with Marco (Q57). Drawn here on the cabin side at the seat plane, where the glass sits on a hexagon.

- Solar cells must stand off the structure: on bare pentagons in full sun the hexagon-pentagon difference could reach about 100 K (about 380 kN on a corner); on a blanket it is 2 to 15 K (1 Nov model).

- The greenhouses get no daylight: 6 to 9 kW of LED per TRUTH 4; the arrays and batteries for that are not sized.

Pentagon frame: P-06 at 180 deep with the 50 mm lip (pentagons carry no glass). Main ribs P-04 (five), hub P-05 with five arms (variant of th
Corner blocks CB-02 (108 deg, tie rows z 128.3 / 364.4). Hardpoints as RF-07. Mitre 16.472 deg; joint-face features by z' (RF-05 table).
The parts library carries the pentagon assembly A-04 and the closure triangle PL-01 (PROPOSAL, 8 Nov).

HCSA RF-09 Pentagon panel, general arrangement, roles, docking	
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: ribs and frame decided; closure plate side OPEN (Q57); roles PROPOSAL (25 Oct)	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

3. The test pieces

These are the parts a shop quotes and makes this winter. The five TP sheets follow this page, and the STEP files for every piece are in the zip that goes with the package (HCSA-Rev-F-test-pieces-STEP-9-Nov-2026.zip, also at hcsa-site.vercel.app/downloads). They were built from the Rev F parameter file the same way the parts library was, so the seam in the test piece is the seam on the flight sheets: 180 mm deep, glands 7.1 x 4.0 at z' 18 and 32, the RTV recess at z' 42, the bolt row at z' 58 on a 75 mm pitch.

What the pieces are

Article	What it is	Pieces	Material	Mass
T1-075	A 75 mm slice of a seam: one bolt, the two cords and the RTV bead, pulled apart in the 100 kN frame	Frame A, frame B, 2 pull blocks	Frames 6061-T651, blocks 4140 QT	3.0 kg of frames, 14.4 kg of blocks
T1-150	A 150 mm slice, two bolts	The same	The same	6.1 and 29.5 kg
T1-225	A 225 mm slice, three bolts	The same	The same	9.1 and 44.4 kg
SA-550	The 550 mm article: a seam with a dummy hardpoint, three 40 mm pins at 60 mm pitch in a 200 mm block, and 2 + 2 seam bolts beside it. The pins are loose by design and the article is pulled in prying through the lugs, so the seam bolts and the seals carry it	Frame A, frame B with the hardpoint block, 2 pull blocks	Frames 6061-T651, blocks 4140 QT	21.9 kg of frames, 109 kg of blocks
SA-550 moved	The same article with the dummy fitting moved onto the line of the pull (z' 46 to 106 instead of 110 to 170). This is the one we make: Rick agreed the moved fitting on 10 November. Sheet TP-03 draws both positions and the STEP zip has both	The same	The same	The same
F-01 to F-03	The pinned fixture that puts the article in the frame: the lug plate, the fork with its 20 mm pin, and the adapter to the machine	3 pieces	4140 QT	See TP-04

Each frame has legs 200 mm long so the pull block can bolt to it from both bearing faces with M12 screws. Run 1 pulls straight through the block centre at z 90 with an M24 rod, run 2 pulls through a pin on the z 24 line, near the outer flange, which pries the seam open the way pressure does. The pull blocks are tall (160 mm) because the seat lip at the cabin face moved the frame to 180 mm deep; the old problem of a 2 mm wall round the M24 thread is gone.

The tolerances, tight only where the seal or the fit needs it

Marco's point on 7 November was that our 0.05 mm everywhere would make every part a precision part. The table on TP-00 proposes two classes. Tight: gland depth 4.0 plus or minus 0.05, because that sets the squeeze on the cord; gland width 7.1 plus or minus 0.10; gland position plus or minus 0.15; the joint face flat to 0.10 over 150 mm and 0.15 over the 550; the three 40 mm bores at plus 0.10 and minus nothing. Normal: the bolt holes drilled

8.5, positioned to plus or minus 0.2; the leg holes 14.0, positioned to plus or minus 0.2 on the pattern; the block faces parallel to 0.10; and ISO 2768-m for everything else. The 7 degree swing of a shop floor moves 550 mm of aluminium by about 0.09 mm, so the measurements that matter are taken at 20 degrees.

Why only two features are tight

The cord is 5.33 mm in a 4.0 mm gland, a squeeze of 25 percent. What eats that squeeze can be added up.

Stack	Gap added	Squeeze left
Nominal cord and gland	none	25.0 %
Thinnest cord (5.20) in the deepest gland (4.05), seam open 0.11 mm under load, each face flat to 0.05	0.21 mm	18.1 %
The same with each face flat to 0.10, the TP-00 proposal	0.31 mm	16.2 %
The same with each face flat to 0.20	0.51 mm	12.3 %

Relaxing the flatness from 0.05 to 0.10 costs two points of squeeze. Nothing else on the frame is in this sum, which is why nothing else is tight. Parker's minimum squeeze for this cord is asked and not yet answered; the leak test on the 550 article with a 0.1 and a 0.2 mm shim under one face would measure it.

Hardware

The test pieces use catalogue hardware from McMaster-Carr. At 4 kN the seam bolt is far below the strength of any 5/16-24 bolt, so its stiffness is what matters, and alloy steel and A286 are equally stiff to within one percent. An alloy steel 5/16-24 screw and nut stands in for the NAS6705 in every test. The flight parts list keeps the NAS numbers.

Two options for the shop

- The blocks in 6061-T651 instead of 4140. On the 550 article each block is then 19 kg instead of 55, and aluminium is quicker to machine. The M12 threads then want inserts.
- Legs 120 mm instead of 200 on the slices. The 150 mm slice with short legs is in the STEP zip as T1-150-DFMlegs120, so a shop can quote both.

What a shop needs from us

The STEP files and the TP sheets, the quantities (one of each slice, one 550 article, two blocks per article, one fixture set), the material and heat treatment, the tolerance table, and the surface finish on the joint face and the glands, Ra 0.8. This zip is what goes to Xometry for instant quotes, as Manos told Marco.

Still open on the test pieces

- Settled on 10 November: we cut SA-550 moved, with the dummy fitting on the line of the pull.
- Whether run 1, the straight pull, stays in the plan now that run 2 pulls at the right depth.

- The leak test on the 550 article: the pass rate, and the port and fittings for it.
- The M24 pull in run 1 is 14 mm off the pull line of the slices (z 90 against z 76); we have noted it and think it does not matter for a straight pull, but it is a question for Marco.
- The T0 glass bars and the T2 quarter panel are not drawn at Rev F yet. The bars follow the ENGN 1440 four-point method, 1 inch by 1/8 inch, and their drawing and arris size are still owed (Q04); the quarter panel waits on the pane clamp.

HCSA test pieces, Rev F, 8 November 2026

The T1 seam slices (75, 150, 225 mm), the 550 mm seal article with its grip legs and dummy hardpoint, and the pull blocks, all at the 180 mm frame.
Drawn from hcsa_params_revF.json (5 Nov). What changed from the 24 Oct Rev E draft: frame depth 140 to 180, gland 7.1 × 4.0, seam bolt NAS6705U27 at 4 kN,
the pull blocks grow to 160 tall to span the legs, run 2 pulls through pinned lugs (RE-03 rev B), the 550 article has legs and a dummy hardpoint (2 + 2 bolts).

Sheets

TP-00	Cover, notes, tolerances, parts list
TP-01	T1 frame B (glands): section, joint face, legs; 75 / 150 / 225 table
TP-02	T1 frame A and the slice as assembled, pull lines, screws
TP-03	550 mm seal article with legs and the dummy hardpoint, both fitting depths
TP-04	Pull blocks (rev B interface) and the lug, fork and pin F-01 to F-03
STEP	out/step: one file per part; out/asm: one assembly per article

General notes

1. Material: frames 6061-T651 plate (as drawn, keeps the FEA twins valid). Pull blocks 4140 quenched and tempered 28-32 HRC (as drawn).
   DFM option for Marco: blocks in 6061-T651 (one material, 65 % lighter: 19 kg instead of 55 kg each on the 550 article).
2. Machining order for the frames: rough, rest 24 h, finish; the joint face is the last cut (fly-cut), so the part is flat after the stress is out.
3. Inside corners web-to-flange and web-to-lip R6. Every hole 1 × 45 deg at both ends. All other edges 0.5 mm break. Gland edges deburred only, no break.
4. Surface finish: joint face and gland faces Ra 0.8 um. Elsewhere Ra 3.2.
5. Hardware (test articles, McMaster-Carr or equal): ISO 4762 M12 × 35 class 12.9 socket screws + ISO 7089 hardened washers into the blocks;
   M12 × 45 class 12.9 for the lug plates; seam bolts NAS6705U27 with KFN670 nuts and two hardened washers (flight part, A286).
6. Seals: Parker BA500-70 butyl cord 0.210 in, two per article, RTV566 bead with SS4155 primer. End dams on the slices (M8 rod, fabrication plan 2.3).
7. Quantities: one of each slice (75, 150, 225) and one 550 article: 8 frames, 8 blocks. Fixture parts F-01 to F-03 are shared: 2 lugs, 2 forks, 2 pins + 1 spare each.

Tolerances: tight only where the seal or the fit needs it (proposal for Marco, 17 Nov)

Feature	As drawn 24 Oct	Proposed	Why
Joint face flatness	0.05 over the length	0.10 per 150 mm, 0.15 over 550	0.1 mm of gap moves the squeeze 2 points of 25 %
Joint face finish	Ra 0.8	Ra 0.8 (keep)	Parker face-seal finish
Gland depth 4.0	+-0.05	+-0.05 (keep)	sets the 25 % squeeze; a shallow groove, easy to hold
Gland width 7.1	+-0.05	+-0.10	fill 78.7 % moves 1 point per 0.1 mm
Gland position z 18, 32	+-0.05	+-0.15	the cords only have to meet the flat face of frame A
Seam bolt holes	8.4 reamed, pos +-0.2 (old)	8.5 drilled, pos +-0.2 (as modelled)	a clamp bolt, not a shear pin, in the test articles
Leg holes	13.5 (old)	14.0 drilled, pattern +-0.2 (as modelled)	M12 screws need play to find the block threads
Hardpoint bores (550)	40 H7	40 +0.10 / 0	the dummy pins are fitted loose by design
Block bearing faces	parallel 0.05	parallel 0.10, height 160 +-0.1	the legs must not be sprung when the screws go in
Everything else	ISO 2768-f	ISO 2768-m	legs, lengths, block outline: nothing seals on them

Temperature: all sizes at 20 C. 7 C of difference moves 550 mm of aluminium by 0.09 mm (23.6e-6 per K), so the long parts are measured at 20 C (Marco, 7 Nov).

Parts list (masses from the solids)

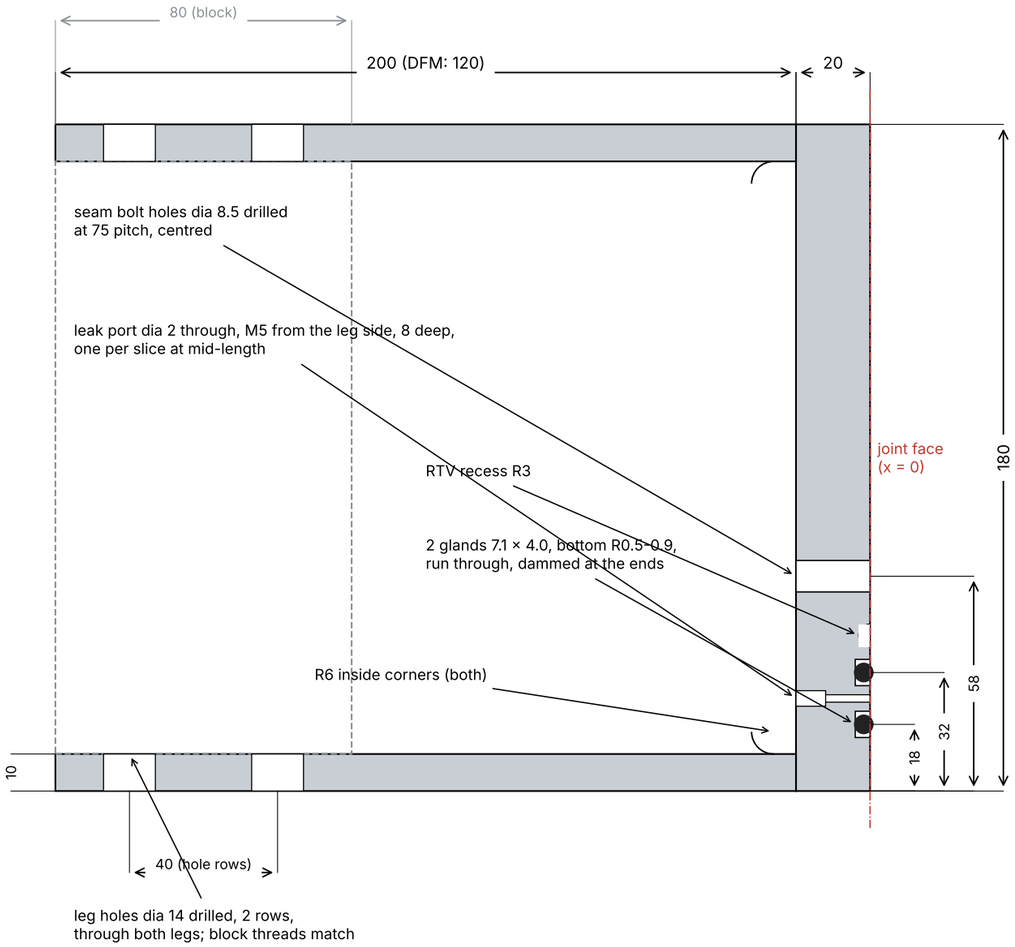
Part	Description	Material	Qty	Mass each kg	Stock blank (mm)
T1-075-FB	Frame B (glands, RTV recess, leak port), 75 mm, 180 deep, legs 200	6061-T651	1	1.49	190 × 230 × 81
T1-075-FA	Frame A (flat joint face), 75 mm, 180 deep, legs 200	6061-T651	1	1.51	190 × 230 × 81
T1-075-PB	Pull block 160 × 80 × 75, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	7.22	165 × 85 × 81
T1-150-FB	Frame B (glands, RTV recess, leak port), 150 mm, 180 deep, legs 200	6061-T651	1	3.02	190 × 230 × 156
T1-150-FA	Frame A (flat joint face), 150 mm, 180 deep, legs 200	6061-T651	1	3.04	190 × 230 × 156
T1-150-PB	Pull block 160 × 80 × 150, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	14.75	165 × 85 × 156
T1-225-FB	Frame B (glands, RTV recess, leak port), 225 mm, 180 deep, legs 200	6061-T651	1	4.52	190 × 230 × 231
T1-225-FA	Frame A (flat joint face), 225 mm, 180 deep, legs 200	6061-T651	1	4.57	190 × 230 × 231
T1-225-PB	Pull block 160 × 80 × 225, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	22.21	165 × 85 × 231
SA-550-FB	Frame B (glands, RTV recess, leak port), 550 mm, 180 deep, legs 200	6061-T651	1	10.87	190 × 230 × 556
SA-550-FA	Frame A (flat joint face), 550 mm, 180 deep, legs 200	6061-T651	1	10.98	190 × 230 × 556
SA-550-PB	Pull block 160 × 80 × 550, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	54.54	165 × 85 × 556
F-01	Lug plate 80 × 100 × 20 (RE-03 rev B)	4140 QT	2	1.05	from the parts library STEP
F-02	Fork with M24 × 3 for the Instron adapter	4140 QT	2	1.80	parts library STEP
F-03	Pin 20 h6 × 70 with shoulder and E-clip groove	17-4PH H900 or 4140 QT	2	0.20	parts library STEP
SA-550-HPmoved-*	The same three 550 parts with the dummy fitting bores at z 76 instead of 140 (make these: moved fitting agreed 10 Nov)	as above	-	same	same

HCSA TP-00 Cover: sheet list, notes, tolerances, parts list

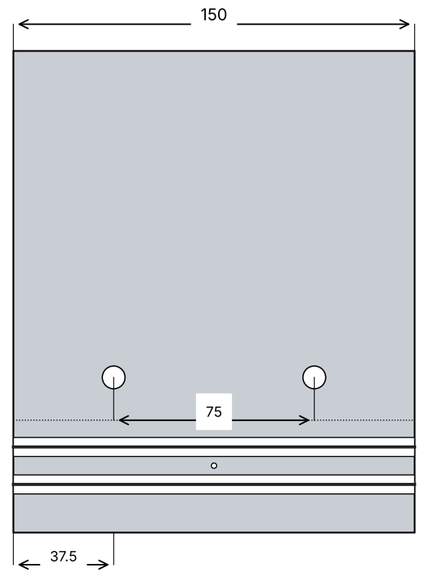
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: for quotation; tolerances and DFM items are proposals for Marco, 17 Nov	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Section through a bolt station, looking along the seam (1:1 on A3)

space side (z 0) at the bottom, cabin side (z 180) at the top; the seat lip (z 170-180) doubles as the second grip leg



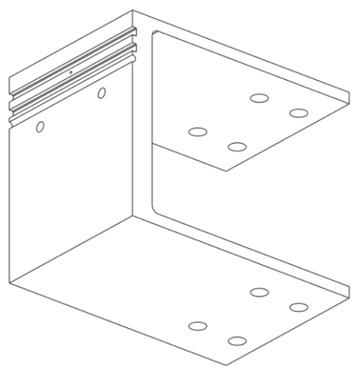
Joint face of the 150 slice (frame B)



Leg holes (both legs, through), measured from the slice end along the seam and from the joint face across the leg

Slice	Seam bolts (from the end)	Leg holes, inner row x 140	Leg holes, outer row x 180	Threads / face
75	37.5	18.75, 56.25	16.5, 58.5 (OPEN-6, moved for the M24 wall)	4
150	37.5, 112.5	37.5, 112.5	37.5, 112.5	4
225	37.5, 112.5, 187.5	37.5, 112.5, 187.5	37.5, 112.5, 187.5	6
550 (TP-03)	50, 125, 425, 500	7 at 75 pitch, 50 to 500	same	14

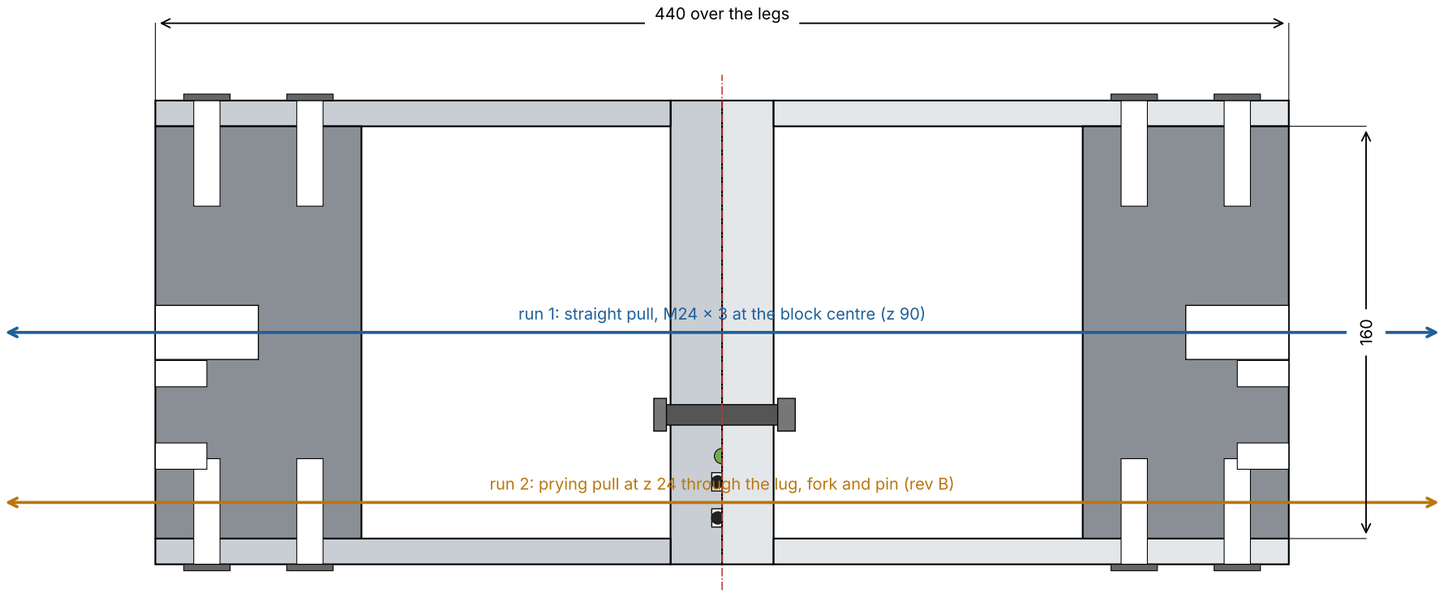
T1-150-FB, from the STEP (legs toward the viewer, joint face at the back)



Frame A (TP-02) is the mirror image without the glands, the recess and the port.
Stock 6061-T651 plate 190 × 230 x (L + 6); with 120 mm legs, 190 × 150 x (L + 6).
The slice is flat, so z equals the joint-plane coordinate z' of the flight frame: the test measures the same joint face the station has.

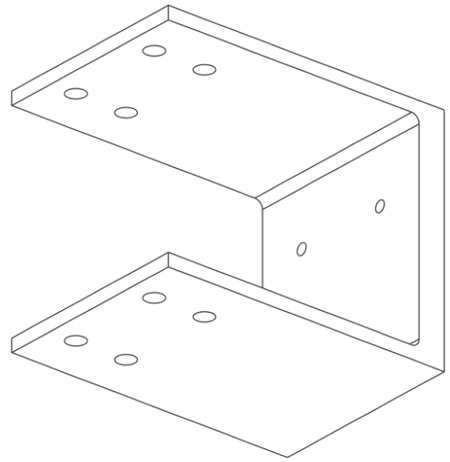
HCSA TP-01 T1 frame B (glands), 180 deep: section, joint face, legs	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: geometry from build_testpieces.py; 75 / 150 / 225 as a table	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Section through a bolt station: frame B (left, glands), frame A (right), the two pull blocks and the seam bolt



M12 × 35 class 12.9 screws with hardened washers through the 14 mm leg holes into the block threads (4 to 14 per face).
The screws carry the pull in shear; the washers keep the head pressure on the 6061 leg under 132 MPa (RE-01-2 check).

T1-150-FA (flat face), from the STEP

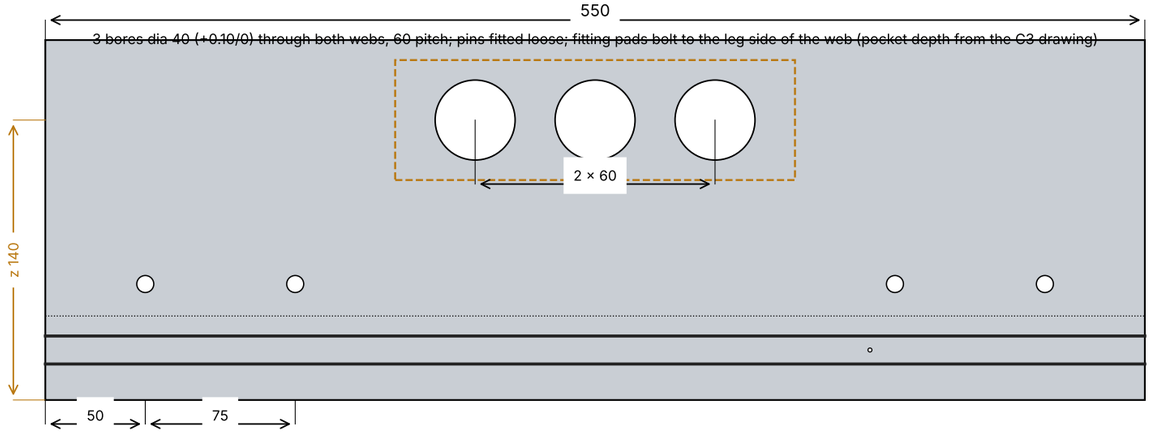


Pull block, 160 × 80 × 150 (TP-04)

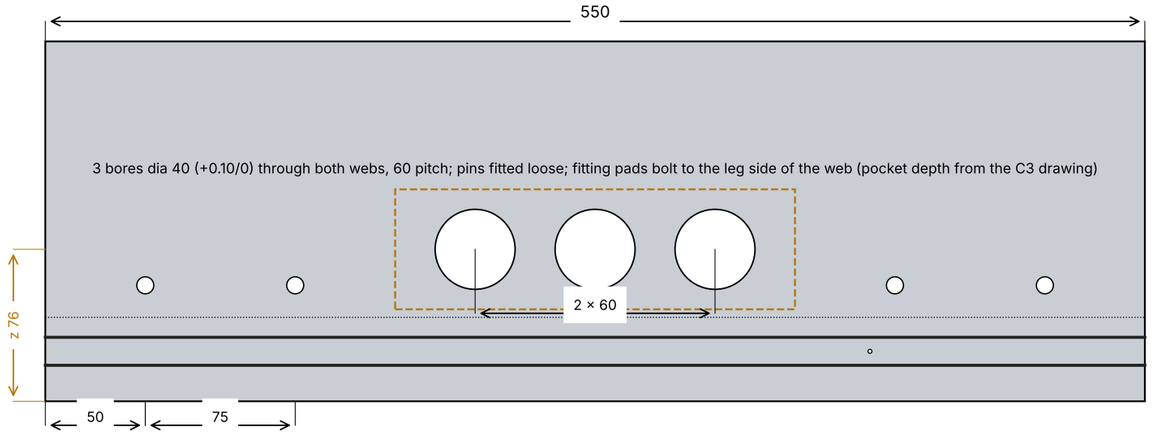
Why the block grew. On the 140 frame the block spanned the flange and the lip, 83 mm apart. On the 180 frame the lip is at the cabin face (z 170-180, P-06), so the span is 160 mm. The single M24 for run 1 sits at the block centre, which puts the straight pull at z 90; the D1 and HP1 twins pull at the frame middle (76 mm), so run 1's pull line is 14 mm off the twin's. Run 2 (the one the seal design turns on) is unchanged at z 24.
The 2.0 and 3.5 mm walls round the two M24 threads of the 24 Oct blocks are gone: rev B has one M24 and the z 24 pull goes through the lugs.
DFM questions for Marco: block in 6061-T651 instead of 4140 (19 kg instead of 55 kg for each 550 block; the M12 and M24 threads in aluminium carry the 100 kN frame load with margin); legs 120 instead of 200 (stock 190 × 150 instead of 190 × 230; the block moves in; the twins' fixture legs change).

HCSA TP-02 T1 frame A and the slice as assembled, pull lines, screws	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: block 160 tall; run 1 M24 at z 90; run 2 through the lugs at z 24 (rev B)	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

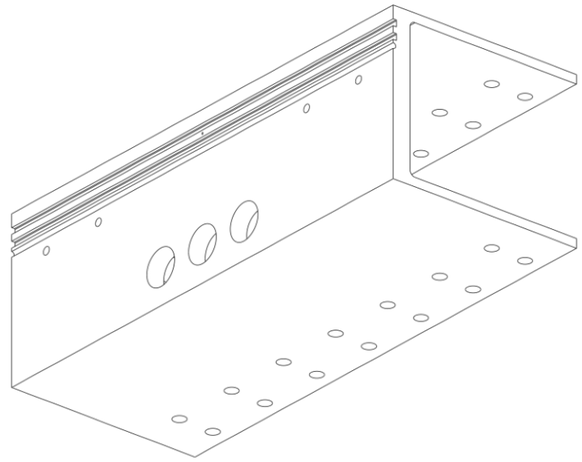
Joint face of frame B, dummy hardpoint as drawn (RE-06-1: fitting 110 to 170 deep)



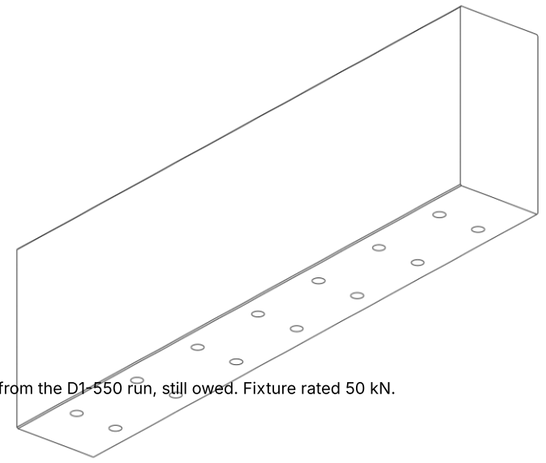
Joint face of frame B, dummy hardpoint moved onto the pull line (HP1M study: 46 to 106 deep)



SA-550-FB from the STEP: legs, 14 + 14 leg holes, 4 seam bolts, 3 pin bores



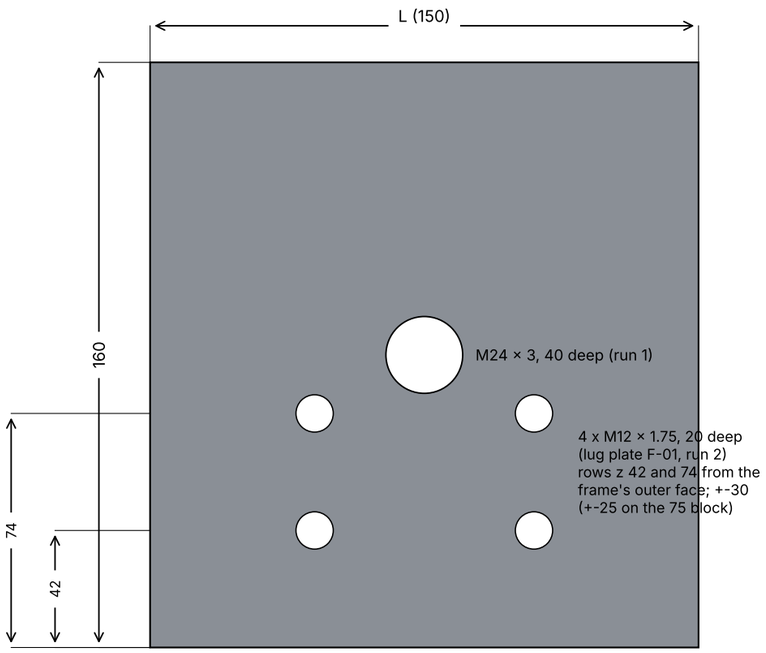
SA-550-PB, 160 × 80 × 550



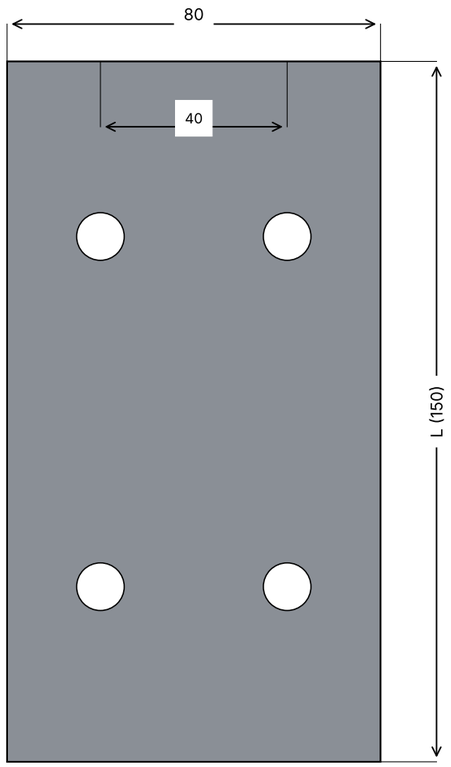
What the article does (RE-03 rev B, 2 Nov): assembled dry, the faces checked with a 0.05 feeler; cords in, bolts tightened from the middle out, squeeze read against torque; bead cured 24 h; end dams fitted; the space between the cords pressurised to 1 and 2 atm and logged (LC5, 7-14 Dec). Then pulled in prying through the lugs while the leak rate is read against the gap, which measures the gap at which the butyl stops sealing, the number the whole seal design turns on. Afterwards it stays assembled as the ageing sample.
Bolts: 2 + 2 at 75 pitch in the two 175 mm bays beside the 200 mm dummy fitting (RF-03), none inside it. Leg holes: 7 positions at 75 pitch (50 to 500), two rows, both legs.
Pull blocks: 160 × 80 × 550 (TP-04), 55 kg each in 4140; 19 kg each in 6061-T651 (DFM option). The seam model reaches the 0.11 mm gap at 3.75 kN per plain bolt; the pull on the article at that point comes from the D1-550 run, still owed. Fixture rated 50 kN.

HCSA TP-03 550 mm seal article with legs and the dummy hardpoint	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: both positions drawn; make the moved one, z 76 (agreed 10 Nov)	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Pull face (the face away from the joint), z measured from the frame's outer face

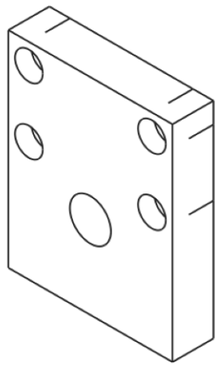


Bearing face (either), the leg screws' threads

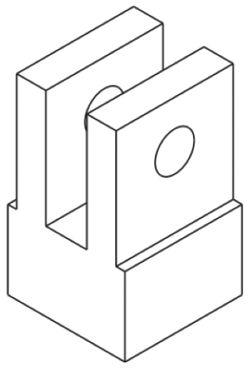


4 x M12 × 1.75 from each bearing face
(10.2 drill 31 deep, 26 full thread),
at the leg hole positions of TP-01.
Holes from the two faces line up;
160 - 2 × 31 leaves 98 mm of steel between.
The outer row sits 20 from the pull face.

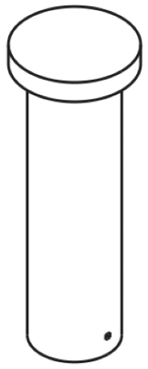
F-01: Lug plate 80 × 100 × 20, 20 H7 pin hole on the z 24 line, 4 x M12 × 45 counterbored



F-02: Fork: two 15 mm cheeks, 21 mm gap, 20 H8 holes, M24 × 3 female for the adapter



F-03: Pin 20 h6 × 70, 28 shoulder, E-clip groove, greased



Load train for run 2: crosshead, adapter, M24 stud, fork F-02, pin F-03, lug F-01, block, legs, frame, joint, and the mirror image below. Both pins parallel to the seam, so each end is free to turn. Checks (RE-03 rev B, 50 kN fixture load): pin double shear 377 kN; lug tear-out 720 kN; worst lug screw 39 kN against an 83 kN proof load. Material 4140 QT 28-32 HRC as drawn.

HCSA TP-04 Pull blocks (rev B interface) and the lug, fork and pin	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: block 160 × 80 x L; M12 from both faces; M24 run 1; lug rows z 42/74; F-01..03 as 2 Nov	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

4. What the computer tests say, Rev F - every run, what it found, and the final picture

Written 7 November, corrected 9 November after an independent check of every number. The corrections are marked where they change a verdict.

This chapter covers every digital test we have run on the Rev F design: the whole station in the shell model, and the corner, the seam, the hardpoint and the pane in Ansys. Nothing here has been tested on real hardware. Every number is from a model, and the glass numbers rest on published data until the bar tests run.

The design the runs describe: main rib option B (I 400 x 100 x 25, web 8), secondary ribs I 180 x 60 x 10, edge frames 180 mm deep, 21 mm BOROFLOAT 33 panes on the 531 mm grid, two butyl loops in 7.1 x 4.0 mm glands plus RTV566, seam bolts NAS6705U27 at 4 kN in one row at z' 58, corner ties at 198 kN. The cabin is 0.8 atm (81 kPa, 26 percent oxygen); the design pressure is 12.3 psia, set by our own relief valve; proof is 1.5 times that (1.26 atm) and ultimate 2.0 times (1.67 atm). The seals are asked to stay shut to 1.93 atm, which is the ultimate pressure times a further 1.15 (the seal requirement set on 4 November). The hatch-open case, where the cabin reaches the host station's 15.2 psia, is checked on top. The allowable for 6061-T6 is 145 MPa.

The short version

1. **The structure passes at 0.8 atm, with room.** The worst member is the hexagon main rib at 103 MPa against 145, and 120 MPa with the temperature band and a warm pentagon added. Nothing yields below the 1.67 atm ultimate; the first yield is at 1.96 atm.
2. **The corner passes, and with the hatch open it is right on its limit.** At 0.8 atm the worst tie is 214.5 kN at proof against 227. If the hatch is left open and the cabin rises to 1 atm, it is 226.3 kN, 0.7 kN under the limit. So the 198 kN preload cannot drift upward. Runs on 14 November show it can come down instead: at 180 kN of preload the worst tie at proof is 198.9 kN, and at 160 kN it is 181.5 kN, while the corner faces open 0.57 and 0.60 mm against 0.8. The hatch-open case has not been run at the lower preloads yet.
3. **The plain seam passes.** The seal gap at the worst bay load is 0.070 mm against the 0.11 mm limit, and the load at which it reaches the limit is about 40 percent above the worst bay load. Moving the bolt row to make room for the seal corners does not work, so the bolts stay at z' 58.
4. **The hardpoint as drawn fails, and we know the fix.** The fitting sits 64 mm deeper than the line of the pull, so the two frames hinge about it and the seals open at 0.2 to 0.5 atm instead of 1.93. Moved onto the line of the pull, the seal at the hardpoint itself holds past 2.5 atm, but the seal band at the plain bolt beside it reaches the limit at 1.87 atm, just short of the 1.93 atm required, so the fix is not complete until that bolt is dealt with. Two runs to settle it were set up on 9 November and restarted on 13 November after the laptop shut down. The first finished that night: moving the plain bolt closer to the fitting (115 mm from its centre instead of 137.5) makes it worse, 0.11 mm at about 1.76 atm.

The second, with that bolt at 6 kN instead of 4, finished on 14 November and just clears it: 0.11 mm at about 1.95 atm. So the fix is a higher preload on the bolts beside each hardpoint, set by measured stretch, because the margin is about 1 percent. The fitting in that run is a spring between rigid pads, not the real pins. Rick agreed the move on 10 November, and it changes the hardpoint sheet.

5. **The panes hold under the steady launch loads.** Even a snug clamp holds the 21 mm pane at 6 g. That is the easy case. A first hand estimate of random vibration (part 6, and the note of 9 November) puts 6.6 to 10.4 kN on a clamp that holds 9.1 kN, so vibration is what sizes the clamp, and it is not yet run in a model.
6. **Leak before break is gone.** The metal yields at 1.96 atm, and the seals in the plain bays open later, at about 2.9 atm. (Beside a moved hardpoint the seal band reaches its limit at 1.87 atm, so there the order is not yet settled.) So the warning comes from the relief valve, the pressure control and the leak ports, not from the seals.
7. **The glass is the one number we do not have.** 21 mm at 0.8 atm passes if the glass's Weibull m is at least 7.0 on our route and 7.7 NASA-style. Only the bar tests can give that number.
8. **Two requirements came out of the checks:** the main rib must be braced sideways at every secondary rib crossing (unbraced it buckles at 0.79 of its load), and the panes should be packed or clamped so launch cannot lift them off their seats.

The table

What	Result	Limit	Verdict	Model, date
Hexagon main rib, 0.8 atm	103 MPa	145 MPa	Pass, 41 % margin	Shell model, 4 Nov
Same, with the ±120 C band and the pentagon 15 K warmer	120 MPa	145 MPa	Pass, 21 %	Shell model, 4 Nov
Hexagon frame, same loads	102 MPa	145 MPa	Pass, 42 %	Shell model, 4 Nov
Secondary rib, same loads	97 MPa	145 MPa	Pass, 49 %	Shell model, 4 Nov
Pentagon main rib, same loads	104 MPa	145 MPa	Pass, 39 %	Shell model, 4 Nov
First yield against ultimate	1.96 atm	1.67 atm	Pass, 17 %	Shell model, 4 Nov
Main rib at proof, no yield	172 MPa	240 MPa	Pass	Shell model, 4 Nov
Hatch open, cabin at 1 atm: main rib, pressure only	127 MPa	145 MPa	Pass, 14 %	Shell model, 4 Nov
Same with the band and warm pentagon added (plus 17 MPa, as in the 0.8 atm row)	about 144 MPa	145 MPa	Pass by under 1 %	Sum, 9 Nov
Frame bow with the band	0.75 mm (0.85 at 1 atm)	1.0 mm	Pass	Shell model, 4 Nov
Main rib sideways buckling, braced every 562 mm	8.4 times	1.0	Pass if braced	Hand calc, 4 Nov

What	Result	Limit	Verdict	Model, date
Same, unbraced over 2.25 m	0.79 times	1.0	Fail: bracing is now a requirement	Hand calc, 4 Nov
Worst corner load	235 kN at the 12.3 psia design pressure, 290 kN at the host's 15.2 psia			Shell model, 5 Nov
Corner tie at proof, 0.8 atm	214.5 kN	227 kN	Pass	Ansys C1, 5 Nov
Same, ties preloaded to 180 kN	198.9 kN; faces open 0.574 mm	227 kN; 0.8 mm	Pass, 28 kN to spare	Ansys C1, 14 Nov
Same, ties preloaded to 160 kN	181.5 kN; faces open 0.604 mm	227 kN; 0.8 mm	Pass, 45 kN to spare	Ansys C1, 14 Nov
Same, pentagon 15 K warmer	213.3 kN	227 kN	Pass	Ansys C1, 5 Nov
Corner tie at proof, hatch open at 1 atm	226.3 kN	227 kN	Pass by 0.7 kN	Ansys C1, 5 Nov
Corner faces open at proof	0.55 mm (0.74 at 1 atm)	0.8 mm	Pass	Ansys C1, 5 Nov
Corner block stress, 95th percentile, at proof	236 MPa (255 at 1 atm)	none set	Open	Ansys C1, 5 Nov
Plain seam, gap at the worst bay load (2.67 kN per bolt), bolt row z' 58	0.070 mm	0.11 mm	Pass, about 40 %	Ansys D1, 5 Nov
Plain seam, bolt row moved to z' 70	0.123 mm at 2.67 kN per bolt (0.111 mm at 2.48 kN, the 1.93 atm load)	0.11 mm	Fail: bolts stay at z' 58	Ansys D1, 5 Nov
Plain seam, pressure where the gap reaches 0.11 mm	about 2.9 atm (3.75 kN per bolt)	1.93 atm	Pass	Ansys D1, 5 Nov
Hardpoint seal as drawn, frames free to turn	0.11 mm at 0.22 atm	1.93 atm	Fail	Ansys HP1, 6 Nov
Hardpoint seal as drawn, frames held at the grips	0.11 mm at 0.47 atm	1.93 atm	Fail	Ansys HP1R, 7 Nov
Hardpoint seal with the fitting moved onto the pull line	0.064 mm at 1.93 atm, under 0.11 to 2.5 atm	0.11 mm	Pass (study)	Ansys HP1M, 7 Nov
Seal band at the plain bolt next to the moved fitting	0.11 mm at 1.87 atm	1.93 atm	Short by 3 %	Ansys HP1M, 7 Nov
Same, plain bolt moved to 115 mm from the fitting centre	0.11 mm at about 1.76 atm	1.93 atm	Worse: a closer bolt is not the fix	Ansys HP1M, 13 Nov
Same, plain bolt back at 137.5 mm, preloaded to 6 kN	0.11 mm at about 1.95 atm	1.93 atm	Pass, about 1 %	Ansys HP1M, 14 Nov
Hardpoint pin, 40 mm	119 MPa	655 MPa	Pass	Ansys C3, Oct
Hardpoint bore chamfers at 1.5 x proof	194 to 197 MPa on 5 to 7 % of points	160 MPa	Open (Q18); about 19 % lower at 0.8 atm	Ansys C3, Oct

What	Result	Limit	Verdict	Model, date
Pane at launch, 6 g, snug clamp	holds; pane lifts 0.012 mm	clamp holds	Pass	Ansys PR1, 6 Nov
Pane at launch, clamp strip squeezed 0.3 mm	9.1 kN clamp, launch changes it by 124 N		Pass	Ansys PR1, 6 Nov
Glass stress in the pane at launch	0.25 to 0.67 MPa	6 MPa	Pass	Ansys PR1, 6 Nov
Pane mean stress with the band	8.8 MPa (10.8 at 1 atm)	set by m		Shell model, 5 Nov
Weibull m the 21 mm pane needs at 0.8 atm	7.0 our route, 7.7 NASA-style	T0 measures m	Open until T0	Fit of the 16 Oct table, 4 and 5 Nov
Same, with the hatch open	7.9 and 8.8			Same
Launch, 6 g along the axis, ribs as cantilevers from the corners	about 10 MPa	145 MPa	Pass (stand-in loads)	Hand calc, 4 Nov
Hexagon-to-pentagon temperature difference	2.3 to 4.6 K; bad cases about 14 to 15 K	15 K design case	At the limit in the bad cases unless the fans run 20 % faster	Orbital thermal model, 1 and 4 Nov
Relief valve against a stuck 1 inch equalising valve	needs about 5 cm^2		Pass if sized	Orifice flow, 4 Nov
Station structure mass	36.1 t in the shell model; the parts lists add up to more (see part 2)		To reconcile	Shell model, 5 Nov

What each model is, and what it found

The whole station (shell model). A beam-and-plate model of all 32 panels in OpenSees, rebuilt in the cloud on 4 November and checked against the 2 November runs to the digit. It is linear, so its results scale with pressure. Three runs are added for the worst case: the direct run at 12.3 psia, the ±120 C frame band, and a run with every pentagon frame and rib 15 K warmer than the hexagons. Adding them is an upper bound. The worst member is the hexagon main rib at 120 MPa, and no member yields below the 1.67 atm ultimate. The model also gives the loads the detailed models use: 280.7 kN on the worst corner at 14.7 psia (235 kN at the 12.3 psia design pressure, 290 kN at the host's 15.2 psia), 108.7 kN per atmosphere on a hardpoint, 1.29 kN per atmosphere on a plain-bay bolt.

The corner, C1. The Ansys model of the three corner blocks with their twelve Inconel 718 ties, rebuilt to Rev F on 5 November with option B rib stubs and the real option B loads. At 198 kN of preload the worst tie is 214.5 kN at proof at 0.8 atm, 213.3 kN with the pentagon 15 K warmer, and 226.3 kN with the hatch open at 1 atm. The faces open 0.55 mm at proof (0.74 at 1 atm) against the 0.8 mm the corner seal is sized for. The block stress has no limit set yet: 236 MPa at the 95th percentile at proof, 255 MPa at 1 atm. Still simplified: the rib stubs were reshaped rather than remeshed, the washers are bonded (with an 8 percent allowance on block stress), and the pins, dowels and the O10 corner seal are not in it. With the ties preloaded to 180 kN instead of 198 (14 November) the

worst tie at proof is 198.9 kN and the faces open 0.574 mm; at 160 kN it is 181.5 kN and 0.604 mm. So the preload can come down a long way before the faces near their 0.8 mm; the hatch-open case still has to be run at the lower preload before we pick it.

The seam, D1. A slice of the plain seam, 180 mm deep, with the two butyl glands, the bolt at 4 kN, and Parker's measured seal force, pulled in prying as the pressure really loads it. The seam opens by rotating about its cabin-side edge from the first newton; there is no parting point, so the bolt preload only shifts the curve a little. With the bolt row at z' 58 the gap between the two cords is 0.070 mm at the worst bay load of 2.67 kN per bolt, and reaches the 0.11 mm limit at 3.75 kN per bolt, about 2.9 atm. With the bolt row moved to z' 70 (to make room for an R16 seal corner) the gap is 0.123 mm, so that move is ruled out and the seal corners have to be solved at the corner block.

The hardpoint, HP1. A 350 mm slice with 75 mm of plain seam on each side of the 200 mm hardpoint, pulled with the real station load through pinned grips. As drawn, the TP3 fitting sits at 110 to 170 mm deep while the pull comes in at about 76 mm, so the fitting acts as a hinge: the frames swing open about it, the seals on the far side open first, and the gap between the cords reaches 0.11 mm at 0.22 atm with the frames free and 0.47 atm with them held at the grips. The plain bolts next to the fitting carry about 60 kN at 1.96 atm, well above what a 5/16 bolt holds (about 41 to 45 kN; the rating memo is still owed), and the frame yields around their holes from about 0.5 atm. In a study run with the fitting moved up onto the line of the pull (pads at 46 to 106 mm deep) the hardpoint seal gap is 0.064 mm at 1.93 atm and never reaches 0.11 mm up to 2.5 atm; the fitting takes nearly all the pull, the plain bolts stay at 13 kN, and the frames turn ten times less. One place is still marginal: the seal band at the plain bolt next to the fitting reaches 0.11 mm at 1.87 atm, just under 1.93. Moving that bolt in to 115 mm from the fitting centre (13 November) brings the limit forward to about 1.76 atm, so the closer bolt is ruled out; in that run the last 60 mm of the slice beyond the bolt has no bolt, which a real seam would, so it may read a little pessimistic. The run with the bolt at 6 kN instead of 4 (14 November) just clears it: 0.11 mm at about 1.95 atm, with the bolt carrying 9.1 kN at 1 atm and 14.4 kN at 1.9 atm, and the seal at the fitting itself at 0.083 mm at 2.5 atm. So the two bolts beside each hardpoint get the higher preload, set by measured stretch because the margin is about 1 percent. The fitting in these runs is a spring between rigid pads, not pins and bores, so the real part has to be drawn and rerun. The 2 November figures of 2.9 to 3.4 atm for the hardpoint seals counted only the fitting's stretch and are withdrawn.

The pane at launch, PR1. One 21 mm, 531 mm pane on its seat with a proposal clamp (a 25 x 8 mm bar on each edge through a 1.5 mm rubber strip), at 6 g and 2 g along the launch axis both ways and 2 g sideways. The pane weighs 56 N as an uncut triangle (54 N as cut), so 6 g is 336 N in total, about 0.2 N per mm of edge. A snug clamp holds it at every load; with the strip squeezed 0.3 mm the clamp carries 9.1 kN and launch changes that by about 1.4 percent. The glass sees 0.25 to 0.67 MPa. So launch does not size the pane or the clamp; tightening does. Steady loads only: random vibration and the 137.7 dB sound field still need a model, and the clamp itself is still a proposal.

The hardpoint fitting, C3 (October). The fitting is a plate from one frame held between two cheeks on the other by three 40 mm pins in double shear. A 40 mm pin holds easily (119 MPa against 655), and the fitting's stiffness, 2.876 million N/mm, is what HP1 uses. At

1.5 times proof a few percent of the points on the aluminium bore chamfers read 194 to 197 MPa against 160; at 0.8 atm they come down about 19 percent. Accept the local yield, bush the bores or thicken them is still Manos's call (Q18).

The glass. We fitted the 16 October thickness table and took the shell factor from the station model. A 21 mm pane at 0.8 atm passes if the Weibull m of the glass is at least 7.0 on our route and 7.7 on the NASA-style route; with the hatch open it needs 7.9 and 8.8. SCHOTT's published 25 MPa fractile gives a factor of 2.8 at the design pressure. We stopped quoting 35 MPa as the glass allowable on 5 November, because nothing SCHOTT publishes backs it. The only large-area float data has an m of about 6, which is why T0 matters more than any other test.

Thermal and launch. The orbital thermal model of 1 November gives a hexagon-to-pentagon difference of 2 to 4 K pressurised and up to 12 K in bad cases; the 15 K design case covers it. At 0.8 atm the fans move thinner air, so the difference grows to 2.3 to 4.6 K (up to 15 K in bad cases) unless the fans run about 20 percent faster. The launch checks use stand-in load factors from the Falcon user's guide (+6/−2 g along the axis, ±2 g sideways), because the Starship guide does not publish them: the ribs see about 10 MPa at 6 g as cantilevers from the corner blocks. Vibration and acoustics are not modelled, and Marco is right that vibration is the launch case that matters.

The final picture, in one paragraph

At 0.8 atm the Rev F structure holds its pressure with a fifth to a half of its strength to spare, nothing yields before ultimate, the corner passes, the plain seams pass, and the panes hold under steady launch loads. Two things are not settled by the models. The glass passes only if the bar tests find a consistent enough glass (m of 7.7 or more), and the hardpoint has to move onto the line of the pull, with the two bolts beside it at 6 kN, which changes a drawing and needs a rerun with the real fitting. On top of those, three things are not modelled yet: launch vibration and acoustics, the corner block stress limit, and the seal loop corners through the corner block.

What still has to run

1. HP1 with the real pins and bores at the new fitting position, after Fusion draws it, and the C3 fitting model rerun for its stiffness there.
2. The corner at 160 or 180 kN of tie preload with the hatch open (1 atm), to pick the new preload. (The bolt beside the hardpoint is settled: at 6 kN it clears 1.93 atm, 14 November.)
3. The pane under a launch vibration spectrum (modal plus random), once SpaceX gives Starship V3 numbers; otherwise with a stand-in spectrum.
4. A stress limit for the corner block, then the block checked against it.
5. D6, the cold seam, with butyl instead of fluorosilicone; D1-550 with the drawn bolt positions; the pre-cambered shell and the 0.2 mm misalignment case.
6. T0 bars (physical): the Weibull m that every glass number waits on.

Sources: "Does option B at 0.8 atm survive (4 Nov)", "Option B at 0.8 atm (4 Nov)", "Rebuild to Rev F (4-5 Nov)", "Decisions and updates (5 Nov)", "TRUTH Rev F draft (5 Nov)" section 9.5, "The hardpoint seal HP1 (6-7 Nov)", "Pane at launch PR1 (6 Nov)",

"Orbital thermal model (1 Nov)", "Glass at 21 mm (4 Nov)", the HCSA Master Document (27 Oct) chapters 14 to 18 for the C3 and Monte Carlo results. Written 7 November 2026.

5. What is still open, by subsystem

Written 7 November, brought up to date on 9 November: items the 8 November drawings and solids closed are marked done, and the sheet numbers are the Rev F ones.

This chapter pulls together the open questions register (rows Q01 to Q101), the Rev F open items of 5 November, the hardpoint and pane results of 6 and 7 November, Marco's comments of 7 November, and the "what is not yet proven" lists in TRUTH Rev D for launch, plants, air, water, thermal and radiation. Each item says who decides and what it holds up. Items that Marco can help with on 17 November are marked **(Marco)**.

1. What Rick decided on Tuesday 10 November, and what still waits on him

Rick agreed these four on 10 November:

- The 0.8 atm cabin with 26 percent oxygen, docked to a 1 atm commercial station through a hatch-shut transfer lock (Q70, Q97 to Q99).
- Leak before break is withdrawn as a requirement, since in the plain bays the metal yields before the seals open (Q93).
- The safety factors: 2.0 at ultimate and 1.5 at proof on the 12.3 psia design pressure, 145 MPa allowable on 6061-T6 (Q68).
- The hardpoint fitting moves onto the line of the pull (the HP1M study). This changes sheet RF-07 and the TP3 envelope.

Still with him: glass bars now on the 0.5 kN frame, or in ENGN 1440 in the spring (Q42); his endorsements for Thornton Tomasetti, Research at Brown and ASTM (Q63 to Q65); and whether the transfer lock rules out the hatch-open case (see corner tie tightening below).

2. Structure

Item	What is open	Who, when	Holds up
Hardpoint fitting	As drawn the hardpoint seal fails (0.22 atm free, 0.47 held, against 1.93). Moved onto the pull line it holds, and Rick agreed the move on 10 November. The nearest seam bolts are 131 to 141 mm from the hardpoint centre, clear of the 200 mm fitting. The fitting is now drawn from the C3 model (HP-02A, HP-02B, A-05): in the 180 mm frame it runs into the outer flange and the secondary ribs, and its pins cannot be put in past the seat lip, so it needs a redesign for the frame	Position decided 10 Nov; Claude redesigns the part; HP1 rerun with real pins and bores	RF-07, the TP3 envelope, the 550 article's dummy hardpoint pockets
Plain bolt next to the hardpoint	Its seal band reaches 0.11 mm at 1.87 atm, just under 1.93. A closer bolt (115 mm from the centre) is worse, about 1.76 atm (13 Nov); a higher preload clears it, about 1.95 atm at 6 kN (14 Nov). How to set 6 kN reliably: measured stretch or a spring under the nut	Manos, Marco	RF-07

Item	What is open	Who, when	Holds up
Corner node, machinability **(Marco)**	Marco: as drawn it cannot be machined; inside corners need radii an end mill can cut; it could be cast, which may change the material. Also the tie-to-fillet clash (Q29), the O10 corner seal section (Q27), the units on Heintges' 4-6 lap (Q28), and the way the main rib flanges enter the block	Manos with Marco, 17 Nov; Fusion	RE-06-2, T3
Corner block stress limit	95th percentile 236 MPa at proof at 0.8 atm, 255 MPa with the hatch open; no limit has been set	Manos, Rick, Henann	Signing the corner
Corner tie tightening	198 kN by measured stretch. The band of 196.5 to 198.1 kN and the 0.7 kN hatch-open margin are both tighter than any tightening method can hold or any model can promise, so either the preload comes down (14 Nov: at 180 kN the worst tie at proof is 198.9 kN, at 160 kN 181.5 kN, faces 0.57 and 0.60 mm against 0.8; the hatch-open case still to run at those preloads) or the hatch-open case is ruled out by the lock. A multi-jackbolt nut would set it with a hand wrench	Manos, with Marco on the method	RF-08, T3
Corner block stock	2219-T852 hand forging was proposed (Q85); we now propose machining from 7050-T7451 or 2219-T851 plate (9 Nov note); no quotes yet (Q58 for ties and titanium)	Manos, suppliers	Cost
Main rib bracing	The star-plate node must hold the main rib's cabin flange sideways every 562 mm (Q95). Drawn as P-11, a proposal with no check behind it	Claude	RF-08, parts A-01
Rib joints	1,620 rib joints inside the pressure boundary and the main rib splice are not designed; the bolted star-plate grid is now the way (Marco, 7 Nov, by serviceability)	Manos	RF-06, cost
Hardpoint bore chamfers	194 to 197 MPa at 1.5 x proof against 160 (Q18): accept, bush or thicken	Manos	Hardpoint drawing
Pane clamp	How the panes are held is a proposal (OPEN-11): the clamp bar, the strip squeeze, the bolt spacing, the 531 or 517 mm cut (OPEN-10), the setting strip hardness (Q13)	Manos	RF-01, RF-06, P-08
Glass	The Weibull m from T0 decides 21 mm and the pane size (708 upside, 425 fallback); SCHOTT still owes flatness, wedge and sag data for 21 mm (Q03, Q06, Q77)	T0 (spring 2027), SCHOTT	Every glass number
Backup pane	A second glass pane or an unloaded acrylic pane behind L7 (Q14); the 1/200 deflection rule (Q15)	Manos, Rick, Henann, after T0	Stack mass
Launch **(Marco)**	Only quasi-static stand-in loads are checked. Random vibration and the 137.7 dB sound field need a model; pack the panels outer face up or rate the clamps (Q96); SpaceX's V3 load factors and spectrum are not public	Claude for the model; Manos asks SpaceX	Packing plan, pane clamps
Flight alloy	Every check in part 4 is against 6061-T6 (145 MPa allowable, 240 yield). Whether the flight frames, ribs and blocks stay 6061 or go to 2219 or 7050-T7451 is open; 7050 is our proposal of 9 Nov	Rick, with Marco	Materials section, the corner block stress limit

Item	What is open	Who, when	Holds up
Shell pre-camber and misalignment	The pre-cambered shell (Q52) and the 0.2 mm misalignment case (Q24) have not run	Claude	T1 run 2

3. Seals

Item	What is open	Who, when	Holds up
Seal loop corners	Parker wants an inside radius of 16 mm at least, 32 ideally; the racetrack end as drawn is R4.3. Moving the bolt row is ruled out, so the loop turns around the end bolt or the flange widens at the corner block. Needs Parker's answer on "two loops" and Fusion (Q92)	Manos, Parker, Fusion	RF-02, the T1 slices, the 550 article, the O10 seal
The sealing gap	0.11 mm is a requirement, not a measurement. Nobody has measured the gap at which butyl stops sealing; the 550 article pulled in prying is what measures it	Test, December to February	The seam requirement wording (Q94)
Butyl life	No stress-relaxation or recovery data for butyl from Parker (Q39); what life test we run	Nobody assigned	TRUTH 10.8
Leak detection	Should each panel have leak detection between the seals (Q38)? With leak before break gone, the ports and the relief valve are the warning	Manos, Rick	TRUTH 14.2
RTV on the ground	Cure the RTV566 bead on the ground in one frame (Q80); venting the outer gap to space against section 6.5 (Q37)	Manos, Rick	Assembly sequence
Seal supply	Two spliced BA500 loops through Sandem or Sealing Specialties; RTV566 sample from Momentive or the SkyGeek kit	Manos, after the funding decision	The 550 article

4. Fabrication and the test pieces

Item	What is open	Who, when	Holds up
A shop **(Marco)**	Blackhawk and JEPIS declined. Marco says Xometry. The STEP files of the frames at 180 mm and the pull blocks exist since 8 Nov	Manos, quotes this week	Everything
Tolerances **(Marco)**	Marco: do not hold 0.05 or even 0.1 mm on every dimension; 7 C of temperature moves 550 mm of aluminium by 0.09 mm; design the part so the machinist does not have to be a hero. The table is on sheet TP-00, and the squeeze sum behind it is in part 3: only the gland depth and the joint face flatness are tight	Marco reviews	The release to the shop
Material **(Marco)**	6061-T651 as drawn, or a 7000-series alloy that needs no heat treatment after machining, as Marco suggests; and the fixturing that holds the part flat. Our position is in part 6	Marco	The RFQ
Design for manufacture **(Marco)**	Marco wants to go through the parts for changes that make them less scary for a shop; the thin walls round the M24 threads in the pull blocks are gone with the 160 mm block	Marco	TP-01 to TP-04
Parts from McMaster **(Marco)**	Spec the test-article hardware from McMaster-Carr; the flight bolts stay NAS parts. The parts memo for NAS6705U27 and its ratings is still owed (Q17)	Manos	The parts order

Item	What is open	Who, when	Holds up
T1 slices and the 550 article	Done 8 Nov as sheets TP-00 to TP-04 and 21 solids; no sheet is checked or released yet (Q55)	Marco's review	The order
Instron adapter	Cannot be drawn until the three frame interfaces are measured with Jerry; the Instron is free from 18 November (Q21)	Manos, Jerry	T1
T0 bars	The test-bar drawing and the arris size from Swift (Q04); two bar sizes and the count (Q51); order after the funding decision	Manos, Swift, Rick	T0
T2 water panel	Redraw the 1:4 article on the Rev F grid with option B ribs and the tie-bolt circle moved in; a host and a safety sign-off (Q45); a budget for B E Peterson (Q46); the gauge map. Marco says do it, with a written plan, video and a process watcher	Manos, Breuer	T2 in February or March
Pentagon closure **(Marco)**	Which side of the ribs the 10 mm closure plate goes on and how it is held and sealed as five triangles (Q57)	Manos, Marco	RF-09
Printed fit checks	PLA was due on 10 November; the PCBWay files still have to be uploaded; the 1:20 model should show option B ribs	Manos	Fit checks
Cold soak	No chamber for the 550 article at -40 C **(Marco)**; D6 not rerun with butyl	Manos	D6

5. Tests and instruments

- Dewesoft loan dates and units, late November (Q44); the spring test plan is due 22 November.
- Trans-Tek LVDTs, yes or no on 12 November (Q48). TML gauges: reply with the 5 m lead wire and the counts (Q47). Chenjie's ENGN 0310 recorder as a backup (Q49).
- Whether a borrowed recorder can sit beside Jerry's frame (Q22); the safety shield; the gauged bolts (not priced).
- The T1 load tables are written per atmosphere at 1 atm and should be restated against 12.3 psia.
- Whether Prof. Henann reviews the test plan (Q50), and the 10 percent pass rule.
- The pass leak rate between the two cords on the 550 article **(Marco)**.

6. Air and fire (life support)

Nobody on this semester's team owns air, water or thermal; Skye and Lauren wrote these sections in the spring. Gerasimos owns the plants.

- At 81 kPa the fans need about 20 percent more airflow to hold the hexagon-pentagon difference, and every fan, avionics box, laptop and medical device has to be checked for cooling at 81 kPa (Q98). Water boils at about 94 C.
- Fire: 26 percent oxygen at 81 kPa is our interpolation inside the ISS fire data; every fabric, cushion, pouch, plant and plastic needs NASA-STD-6001 testing at the real mix.
- From TRUTH Rev D 5.9: the scrubber, fan and isolation hardware are first-order estimates scaled from ISS parts; the 24-hour safe-haven window is a planning assumption with no fault tree; the mode switching (unoccupied, visit, safe haven) is not

specified; the emergency oxygen and nitrogen tanks are not specified; the plant contribution in an emergency is not credited.

- The TRUTH 5.3 setpoints still say 1 atm and ISS practice; they change to 81 kPa and 26 percent oxygen in Rev F.

7. Water and thermal

- From TRUTH Rev D 6.7: the evaporator and condenser pentagons must face sun and shadow, which depends on the orbit and attitude; the ring beam length depends on where the greenhouse pentagons sit; the thermal balance uses guessed emissivity and sink temperature; the inner pane may drop below the dew point in eclipse with the plants transpiring; the distillation loop's start and stop across eclipse is not worked out; the pump is not chosen; there is no backup water path if the loop fails.
- The 3 November research page says the condenser takes up to about 36 kg a day of humidity and a dehumidifier does the rest; the condensation reasoning behind the outer gap being vented to space has to be checked again.
- A time-varying thermal model of the whole station, and who builds it (Q87). The 1 November orbital model covers the hexagon-pentagon difference only.
- Solar cells must stand off the pentagon structure on an insulating blanket, or the hexagon-pentagon difference could reach about 100 K.

8. Power, the pentagons and the column

- Which pentagons carry which roles (eleven identical and one dock, as Manos set on 25 October, against the earlier split in TRUTH and the model), and which way the column runs (Q88).
- Whether the inflatable shield only holds shape or must hold breathable air as a repair balloon; at breathable pressure it pulls hundreds of kN on its pentagon (Q89).
- Stabilising thrusters in or out (Q90).
- The greenhouses get no daylight: 6 to 9 kW of LED, and the LEDs are not qualified for LEO thermal cycling (TRUTH 4.9). The solar array and battery sizing for that load is not in TRUTH.
- The column carries no pressure load and must slide at one end; TRUTH disagrees with itself on where it ends.

9. Docking and the transfer lock

- The transfer lock is a concept, not sized (RF-04): the relief valve of at least 5 cm², the 2.3 to 2.9 mm equalising orifice, who owns the vestibule valves (Q97).
- Whether a commercial station operator (Orbital Reef or another) accepts a hatch-shut interface (Q99); Stellar Amenities is the first contact. If not, the module runs at 1 atm, where the structure passes with 14 percent margin and the corner sits on its limit (Q100).
- From TRUTH Rev D 5.9: each candidate host has a different docking and ECLSS interface; the ISS-era "shared atmosphere" wording has to go.

10. Radiation and the MMOD shields

- From TRUTH Rev D 7.8: no Monte Carlo dose model yet; the 1 to 2 cm HDPE layer is a first-order pick; the deployment mechanism of the pentagon-localised inflatable shields is not designed; the standoff struts and their thermal cycling are not designed; shields at 360 to 630 kg against a 250 kg placeholder; crew dose tracking is operations work.
- The storm shelter was dropped; visitors go back to the host station on a solar particle event warning.

11. Plants

- From TRUTH Rev D 4.9: automated care between visits is not validated at six-alcove scale; plant survival through launch vibration is untested; nutrient dosing reliability, pests, cross-contamination between alcoves, seed viability and crop-loss contingency are open research topics.
- The tasting trays, the drinks ring and the plant placement on the mast (5 November interior) are not in TRUTH yet.

12. Interior

- Marina to confirm the six pods: size, shape, count, and whether the mouths face out or toward the mast.
- Pod shell material and mass, the strut mounts, how the pods come off for launch and service; the fire rating of cushions and pouches at 26 percent oxygen.
- The ramp and cycling track from Marina's images: in or out (Q61). The circulation lengths redone on the Rev F interior.
- The 67-question interior brief of 25 October: construction, sizes, masses, attachment and ownership of every interior piece.
- The front-page loop video still shows the older, larger pods; it needs rendering again on the laptop GPU.

13. Assembly in orbit and operations

- From TRUTH Rev D 3.7: magnetic self-alignment has not been shown at 2.25 m panel scale; the assembly time (70 to 190 hours in different places) is not validated; shirtsleeve assembly inside Starship is not an established practice.
- Zero-clearance pins against magnet-guided alignment to 1 to 2 cm: tapered or expanding pins are a proposal **(Marco)**.
- How the panels are repaired in orbit, which Marco says should drive the frame build; a damaged rib segment is one part to replace in the bolted grid.
- A ground trial: the docking pentagon with its first ring of five hexagons.
- The Starship V3 payload envelope drawing, which we still do not have; the 20 flat layers with the column upright beside them (Q79).

14. Documents, website and the paper

- TRUTH Rev F exists only as replacement text; there is no assembled Rev E or Rev F document. Rev D (14 October) is the last full one. Rick answered Q68, Q70 and Q93 on

10 November, so Rev F can now be assembled and signed (Q76).

- The 47 items on the Rev E fix list of 24 October are not applied. The team pages still show the spring team (Q73); the Heintges credit wording (Q74); the case for the shape (Q69); some meetings have no notes (Q71).
- The Master Document of 27 October is Rev E; this package is its Rev F addendum until it is rebuilt.
- The website carries Rev F since 5 November. Since 9 November the parts library has the corner blocks, the hardpoint fitting, the pentagon closure plate and panel, the node assembly and the pane at 21 mm; it still lacks the shutters and the outer layers, and PL-01 and A-04 still use the straight corner block of 8 November. Marco's credit no longer says "PhD" (13 November).
- A conference entry for any talk other than Space Horizons 2026; a Rev E spacewalk and robot assembly time for TRUTH.

15. Money and schedule

- No purchasing budget until the funding decision comes through. Then: Trans-Tek, the Swift bars, the cord, the RTV and the A286 bolts.
- Thornton Tomasetti ($13,612 for T1), Research at Brown (up to $500) and Champlin ($19,885 for the water rig) are open; the NASA Rhode Island Space Grant is spring 2027.
- The 550 article leak test needs machining to start by mid-November to make 7 to 14 December; the T1 slices are needed before the late-January Dewesoft loan; the fall report is due 11 December.
- The flight cost estimate ($8M to $26M for the structure) has no quotes behind most lines; Marco thinks many of our force values add cost without need.

Sources: Open questions register (4 Nov), TRUTH Rev F draft (5 Nov) open items, What is open (4 Nov), The hardpoint seal HP1 (6-7 Nov), Pane at launch PR1 (6 Nov), Marco's review (7 Nov), Interior update (5 Nov), Conference gap list (3 Nov), the HCSA Master Document (27 Oct) chapters 11, 18, 20 and 24, and TRUTH Rev D sections 3.7, 4.9, 5.9, 6.7 and 7.8 (read from the Drive copy; the text export stops at section 8.3, so sections 8 to 16 were not re-read for this list). Written 7 November 2026.

6. For the call with Marco: four positions and three open questions

After Marco's emails of 7 November we worked each of his points with numbers (the note of 9 November has the sums). So the call can start from positions he can agree with or knock down, and keep the time for the three things we cannot answer ourselves.

Positions: agree or disagree

1. The corner block is machined from plate, not cast and not die forged (RF-08, CB-01). It carries seal loops on two faces, and a casting's porosity is a leak path there; there are only 180 blocks, too few for a die. Cut from a billet it starts at 87 kg and ends at 23 kg. Every surface is a plane or an R10 corner parallel to one line, so one set-up on a rotary axis does the faces, the landing and the holes. The R10 corners are 430 mm long and have to be cut from the open side. The weak spots are the 17 to 20 mm of metal between the rib landing and the joint face, and a washer seat 3.4 mm from the landing edge on the pentagon block.

2. 6061-T651 for every test piece; 7050-T7451 proposed for the flight frame and blocks. The test pieces are only pulled, at low load. For flight, the hardpoint run puts the frame at 235 MPa against a 240 MPa yield in 6061; 7050-T7451 yields near 440, is made thick, and needs no heat treatment after machining. Nothing is welded since the bolted grid was chosen.

3. Only two features are tight on the test pieces (TP-00). The gland depth, plus or minus 0.05, and the joint face flatness, 0.10 over 150 mm and 0.15 over 550. They are the only two that change the squeeze on the cord, which stays above 16 percent in the worst stack. Everything else is ordinary shop tolerance, lengths measured at 20 degrees. Hardware for the test pieces comes from McMaster.

4. The corner ties are set with a multi-jackbolt nut, and the stretch is read on the stud. 198 kN in a 7/8 stud would take 650 to 900 N m on a plain nut. A jackbolt nut does about 145 kN on an M24 thread with eight screws at 14 N m, and 215 kN on M27 with six at 36 N m, so our load needs the high-strength series or a 1 inch stud. The corner runs of 14 November say the preload can come down at 0.8 atm: at 160 kN the worst tie at proof is 181.5 kN, 45 kN under the limit, and the faces open 0.60 mm against 0.8. That also brings it inside what an M27 jackbolt nut does (215 kN).

Open: we need your judgement

5. The hardpoint fitting (RF-07). The position is settled by the models: on the line of the pull. The part is not. It is a titanium lug in a two-cheek clevis with three 40 mm pins along the panel depth, and in the 180 mm frame it runs into the outer flange and the secondary ribs, and its pins cannot be put in past the seat lip. How would you build a pinned joint that carries about 110 kN per atmosphere across a seam, is stiff (it must stretch less than a tenth of a millimetre), can be closed by a crew from inside, and fits in 180 mm of depth? Pins from the side, a tapered or expanding pin, or something that is not a pin at all?

6. Launch vibration. Our first hand model puts the pane's first mode near 600 Hz. With NASA's general component spectrum as an upper bound (0.16 g squared per hertz, 14.1 g rms), the pane pushes 6.6 to 10.4 kN on a clamp that holds 9.1 kN, and the glass sees 3.5 to 5.5 MPa. So the clamp squeeze, not the glass, is what vibration sizes. Is that the right placeholder spectrum, and is a tap test of one clamped pane the right first test?

7. Holding frame B to cut the glands and the joint face in one setting. That is the fixturing question behind the flatness, and it is the machinist's, so we would like your view before the drawings go to Xometry.

If there is time

- The pentagon closure plate: which side of the frame, and fixed how (RF-09, Q57).
- The cold soak: there is no chamber for the 550 article at minus 40 degrees.
- The pass leak rate for the 550 article, and how a shop would pressure test a seam of this kind.
- The seal loop corners at R16, which have to be solved in the corner block (RF-02, Q92).

7. Where everything lives

What	Where
This package, as a PDF, and its one-page summary	Drive, UPDATES / Week 10 (9 to 15 Nov 2026), and on the website, linked from /blueprints/#revf (team password)
The Rev F sheets RF-01 to RF-09 and the test-piece sheets TP-00 to TP-04	Website, /blueprints/#revf, each as an image and as a PDF; Drive, Week 10
The test-piece STEP files, 21 solids and 6 assemblies	Website, /downloads (HCSA-Rev-F-test-pieces-STEP-9-Nov-2026.zip); Drive, Week 10
The parts library, every flight part with STEP, GLB and four views	Website, /parts
The master 3D model, with the taken-apart view	Website, /model (team, behind the password) and /explore (public)
The computer test results, the Rev F decks and the result files	The laptop, Documents\HCSA_FEA\F_revF_4nov; the write-up is part 4 of this package and the Project note "What the computer tests say, Rev F (7 Nov)"
TRUTH, the master spec	Drive, 01 TRUTH, "HCSA TRUTH Rev F (draft, 5 Nov 2026)"; the signed version is still Rev D of 14 Oct
The open questions register	Drive and the Project note "Open questions register (4 Nov)", with the rows that changed on 5 Nov in the decisions note
Marco's review and our answers	The Project note "Marco's review (7 Nov)", its copy in Drive (UPDATES / Week 09), and the email thread of 7 Nov
The code that draws the sheets and builds the solids	GitHub, Manos-Theofanous-ET/hcsa-interactive, and the source zip beside the STEP zip in Drive
The website source	GitHub, Manos-Theofanous-ET/hcsa-interactive; main deploys to hcsa-site.vercel.app through Vercel
The Master Document of 27 October (Rev E, 75 pages)	Drive, UPDATES / Week 8; kept for the record, this package replaces it

The Drive folder for the project is the one shared with the team, with UPDATES by week and the numbered topic folders. Anything made this week goes into Week 10 and into its topic folder.

Appendix. Marco's points worked with numbers (9 Nov)

Marco's review of 7 November raised six things. On 8 November we answered them mostly with drawings. This note answers them with numbers, so the call can start from a position and not from a question. Everything here is a hand calculation or a catalogue figure unless it says it is a model run, and each one ends with what would prove it.

1. Tolerances: what the seal needs, and what it does not

The seal is a 5.33 mm butyl cord in a 4.0 mm deep gland, so it is squeezed 25 percent. Four things eat that squeeze, and they can be added up.

What	Size	Squeeze left
Nominal cord and gland	5.33 in 4.00	25.0 %
Thinnest cord (5.20) in the deepest gland (4.05), the seam open 0.11 mm under load, faces flat to 0.05 each	0.21 mm of gap	18.1 %
The same with faces flat to 0.10 each (TP-00 proposal)	0.31 mm	16.2 %
The same with faces flat to 0.20 each	0.51 mm	12.3 %

So going from 0.05 to 0.10 mm flatness costs two points of squeeze, and the seal still has 16 percent in the worst stack. Going to 0.20 would not be safe without Parker's word on the minimum. That is the case for the split on sheet TP-00: the gland depth (plus or minus 0.05) and the joint face flatness (0.10 over 150 mm) are the only two tight features, because they are the only two in this sum. Everything else on the frame, the bolt holes, the leg holes, the length, the flange, can be ordinary shop tolerance, since none of it changes the squeeze.

What would prove it: Parker's minimum squeeze for BA500 in a face seal (asked, not yet answered), and the leak test on the 550 article with shims of 0.1 and 0.2 mm under one face.

2. Material: 6061 for the test pieces, a 7000-series alloy where the flight frame yields

Marco suggested 7000-series aluminium. For the test pieces the answer is still 6061-T651: they are only pulled, the load is low, every shop stocks it, and T651 plate is stretched, so it moves least when it is cut. The way to keep it flat is in the cutting order, not the alloy: rough both sides, let it rest, then finish the joint face and the glands in one setting.

For the flight frame his suggestion has a real case. The hardpoint run with the moved fitting (HP1M) puts 235 MPa in the frame at 1.9 atm and 291 MPa at 2.5 atm, against a yield of 240 MPa for 6061-T6. A stronger alloy turns that from marginal to comfortable.

Alloy and temper	Yield, MPa	Ultimate, MPa	Notes
6061-T651 plate	240	290	Cheap, everywhere, welds; the test-piece choice

Alloy and temper	Yield, MPa	Ultimate, MPa	Notes
2219-T851 plate	320 to 350	430 to 450	The pressure-vessel alloy, welds, good cold; the flight baseline
7075-T7351 plate	390	475	Strong, the T73 temper resists stress corrosion; does not weld
7050-T7451 plate	440	510	Made for thick sections, keeps its strength through 150 mm and more; does not weld

The values are typical handbook figures and must be replaced by the design allowables before anything is sized. None of these needs heat treatment after machining, which is Marco's other point. The bolted grid was chosen on 7 November, so nothing in the frame is welded, and the usual objection to 7000-series is gone. Proposal for Rick and Marco: keep 6061-T651 for every test piece, and carry 7050-T7451 as the flight alloy for the frame members and the corner blocks, in place of 2219.

3. The corner block: machine it from plate, do not cast it, do not forge it

The block weighs 22.9 kg with its leaning faces. Cut from a 2219 or 7050 billet sawn along the lean it starts at about 95 kg, so three quarters is chips. That sounds wasteful and is still the right answer, for three reasons.

- The block carries seal loops on two faces. A casting has porosity, and porosity on a sealing face is a leak path. A casting also carries a casting factor on its strength, 1.25 or more, on a part that is already near its limit.
- A forging needs a die. There are 180 blocks on the station, 120 of one kind and 60 of the other, and a die does not pay for that number. A hand forging is only a better billet.
- Plate in 7050-T7451 is made thick with its strength guaranteed through the thickness, so the 400 mm depth would be cut along the plate, not through it. Every surface of the block is a plane or an R10 corner parallel to the vertex line, which suits one set-up on a rotary axis.

The stud seat is fine in a wrought alloy. The 45 mm washer on the 23.5 mm hole bears on 1,157 square millimetres, which is 171 MPa at the 198 kN preload and 196 MPa at the 227 kN limit.

A correction to our own count: there are 60 corner nodes but 180 corner blocks, three at each vertex. The 8 November text said 60 blocks in two places. A node is 80 kg, so the 60 nodes are 4.8 t.

What would prove it: a quote for one CB-01 from the STEP file, which Xometry can give this week.

4. Tightening the ties: the force is reasonable, the wrench is not

Marco called several forces achievable but unreasonable. For the corner tie the stress is ordinary: 198 kN in a 7/8-14 stud is 603 MPa, and 691 MPa at the limit, in Inconel 718 that yields above 1,000 MPa. What is unreasonable is the wrench: 650 to 900 N m on a nut, by an astronaut, to within one percent.

There is a catalogue part made for this. A multi-jackbolt tensioner replaces the nut with a ring of small screws that each push on a hardened washer. Nord-Lock's Superbolt MT series reaches 145 kN on an M24 thread with eight screws at 14 N m each, and 215 kN on an M27 thread with six screws at 36 N m. So the tie can be set with a small hand torque wrench. The two questions it raises are size and accuracy: our 198 kN is above the standard M24 nut's capacity of 185 kN, so it needs either the high-strength series or a 1 inch stud, and the scatter of a jackbolt nut is better than a plain nut but not the one percent the hatch-open case wants. The stretch can still be read by an ultrasonic gauge on the stud end, and that reading, not the torque, is the record.

The other half of the answer is to need less force. Two corner runs at 180 and 160 kN preload, at 0.8 atm, finished on 14 November. In both the faces stay under 0.8 mm at proof (0.574 and 0.604 mm), and the worst tie drops to 198.9 and 181.5 kN. So the preload can come down, and so does everything that follows from it; the hatch-open case has to be run at the new preload before we pick it.

5. Launch vibration: a first model that needs no data from SpaceX

Marco said vibration is the hard case and asked for a real model. This is the first rung of one, for the pane, done by hand so that the finite element run has something to be checked against.

- The pane is a 531 mm triangle of 21 mm glass, 5.7 kg. On rigid simple supports its first mode is about 985 Hz. On the two EPDM strips of the proposal clamp it also bounces as a whole at about 780 Hz. Together the first mode is near 600 Hz.
- With no spectrum from the launcher, the usual placeholder is NASA Goddard's general component level (GSFC-STD-7000, GEVS): 0.16 g squared per hertz from 50 to 800 Hz, 14.1 g rms overall, for qualification. It is meant for parts under 22.7 kg bolted to a spacecraft, and the standard lowers it for heavier items, so for a pane sitting on a 1.5 t panel it is an upper bound.
- Miles' equation then gives the pane's response at its own frequency. With a damping factor Q of 10, which is fair for a pane on rubber, it is 39 g rms, so 117 g at three sigma. With Q of 25 it is 186 g.
- At 117 g the pane pushes on its supports with 6.6 kN, the same as 54 kPa of pressure spread over it, and the glass sees about 3.5 MPa. At 186 g it is 10.4 kN, 85 kPa and 5.5 MPa. The cabin pressure itself is 81 kPa and gives 5.2 MPa by the same formula, so the glass is not the problem.
- The clamp is. The 6 November pane model clamps the pane with 9.1 kN at 0.3 mm of squeeze. That holds against 6.6 kN and does not hold against 10.4 kN. So the squeeze on the clamp strip is set by vibration, not by the steady launch load, and the damping of the strip decides whether 0.3 mm is enough.
- Noise acting directly on the pane is small. The Starship guide gives 137.7 dB overall in the payload bay, which is 153 Pa rms, against 81,000 Pa of cabin pressure. Noise matters through the panel it shakes, not through the pane.

What this model does not have is the panel. The pane rides on a 1.5 t ribbed panel stacked twenty deep for launch, and the panel's own modes decide what reaches the pane. That is the next rung: a modal run of one hexagon as packed, then a random response with the

GEVS level at the stack supports. The first test a university lab can do is a tap test of one pane in its clamp, with an accelerometer, to measure the frequency and the damping. Those two numbers replace the 600 Hz and the Q of 10 above.

6. Hardware from McMaster for the test pieces

The test pieces do not need NAS bolts. At 4 kN the seam bolt is at a small fraction of any 5/16-24 bolt's strength, so what matters is its stiffness, and steel and A286 are the same stiffness to within one percent. An alloy steel 5/16-24 socket screw and nut from McMaster stands in for the NAS6705 in every test, with a note on the sheet. The M12 and M24 fixture screws are catalogue parts already. The flight parts list keeps the NAS numbers.

7. The hardpoint: what the check found

The independent check on 9 November found that our 8 November hardpoint solids were wrong, and recovering the real fitting from the October model showed it does not fit the 180 mm frame. That is set out on sheet RF-07 and in part 2, and it is question 5 for the call.

What changes in the package

- Part 3 gets the squeeze table next to the tolerance table, and the McMaster note.
- Part 6 asks Marco to agree or disagree with four positions (plate not casting, 7050 for flight, the tolerance split, a jackbolt nut on the ties) and keeps three open questions.
- Part 5 gets launch vibration as its own item with the first numbers, and the clamp squeeze tied to it.
- The count of corner blocks is corrected to 180 wherever it said 60.

Written 9 November 2026. Sources: Marco Cross's emails of 7 November; the HP1 and PR1 notes of 6 and 7 November; the parts solids of 8 November; GSFC-STD-7000 (GEVS) table 2.4-3 as reproduced by Vibration Research; the Starship Users Guide acoustic table; Nord-Lock's Superbolt MT dimension sheet. Plate properties are typical handbook values, not design allowables.